Local Constraint-Based Sparse Manifold Hypergraph Learning for Dimensionality Reduction of Hyperspectral Image

Yule Duan, Hong Huang, *Member, IEEE*, and Yuxiao Tang

***Abstract*— Sparse representation-based graph embedding methods have been successfully applied to dimensionality reduction (DR) in recent years. However, these approaches usually become problematic in the presence of the hyperspectral image (HSI) that contains complex nonlinear manifold structure. Inspired by recent progress in manifold learning and hypergraph framework, a novel DR method named local constraint-based sparse manifold hypergraph learning (LC-SMHL) algorithm is proposed to discover the manifold-based sparse structure and the multivariate discriminant sparse relationship of HSI, simultaneously. The proposed method first designs a new sparse representation (SR) model named local constrained sparse manifold coding (LCSMC) by fusing local constraint and manifold reconstruction. Then, two manifold-based sparse hypergraphs are constructed with sparse coefficients and label information. Based on these hypergraphs, LC-SMHL learns an optimal projection for mapping data into low-dimensional space in which embedding features not only discover the manifold structure and sparse relationship of original data but also possess strong discriminant power for HSI classification. Experimental results on three real HSI data sets demonstrate that the proposed LC-SMHL method achieves better performance in comparison with some state-of-the-art DR methods.**

***Index Terms*— Dimensionality reduction (DR), discriminant sparse hypergraph, hyperspectral image (HSI), manifold learning, sparse representation (SR).**

I. Introduction

BENEFITING from the high spectral resolution of remote sensors, HSI captures dozens to hundreds of contiguous spectrum bands of ground objects and provides rich spectral information for classification [1], [2]. However, HSI usually contains a large number of redundant information that will weaken the performance of land cover classification [3]. The high-dimensional characteristic of HSI data also makes classifiers suffer from the Hughes phenomenon caused by the small training sample size, which brings a huge challenge to the practical application of HSI [4], [5].

Dimensionality reduction (DR) is an applicable way to overcome the drawbacks of HSI, and a large number of DR methods have been proposed [6], [7]. Manifold learning as a popular DR method is able to reveal the nonlinear characteristics of data. Locality preserving projections (LPP) and neighborhood preserving embedding (NPE) as two classical manifold learning methods are able to effectively reveal the nonlinear structure of high-dimensional data [8], [9]. LPP learns low-dimensional features by preserving the local geometrical structure of data, while NPE applies the Laplacian matrix to maintain the local neighborhood relationship between embedding data points. Marginal Fisher analysis (MFA) is a supervised manifold learning method, and it constructs intraclass and interclass graphs to characterize the discriminant manifold relationships of high-dimensional data [10]. On the basis of MFA, a local geometric structure Fisher analysis (LGSFA) method was proposed to further consider preserving local geometric structure when maintaining discriminant relationships [11]. However, these manifold-based DR methods reveal the geometrical structure generally rely on the Euclidean distances that are sensitive to data noise [12]. Moreover, they mainly focus on the local manifold structure but hard to reveal the real subspace structure of high-dimensional data.

With the advancement of compressed sensing, sparse representation (SR) has become an effective mathematical tool that reveals the subspace structure of data and selects sparse neighbors adaptively [13]–[15]. Therefore, some SR-based DR methods were developed for dealing with high-dimensional data. Sparsity preserving projections (SPPs) and discriminative learning by SR projections (DLSP) are both classical sparsity-based DR algorithms. SPP constructs a ℓ_1-graph to maintain the sparse reconstructive relationship of data [16], while DLSP incorporates the discriminant geometrical structure and sparsity of SPP to extract embedding features [17]. In addition, sparse and low-rank graph-based discriminant analysis (SLGDA) is presented for learning low-dimensional features of data by integrating SR and low-rank representation into graph embedding framework [18].

However, these SR-based methods are not suitable for HSI data because they generally assume that data points can be

Manuscript received February 25, 2020; revised April 16, 2020; accepted May 15, 2020. Date of publication May 29, 2020; date of current version December 24, 2020. This work was supported in part by the Basic and Frontier Research Programmes of Chongqing under Grant cstc2018jcyjAX0093, in part by the Innovation Program for Chongqing's Overseas Returnees under Grant cx2019144, in part by the Chongqing University Post-Graduates Innovation Project under Grant CYB18048, and in part by the Fundamental Research Funds for the Central Universities under Grant 2019CDYGYB008. *(Corresponding author: Hong Huang.)*

The authors are with the Key Laboratory of Optoelectronic Technology and Systems of the Education Ministry of China, Chongqing University, Chongqing 400044, China (e-mail: duanyule@cqu.edu.cn; hhuang@cqu.edu.cn; tangyuxiao@cqu.edu.cn).

Color versions of one or more of the figures in this article are available online at https://ieeexplore.ieee.org.

Digital Object Identifier 10.1109/TGRS.2020.2995709

0196-2892 © 2020 IEEE. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 03:23:59 UTC from IEEE Xplore. Restrictions apply.

embedded in the ambient linear subspace [19]. In practical situations, HSI suffers from the nonlinear disturbance in the imaging process and exhibits significant nonlinear structure [20], [21]. Fusing manifold learning into the SR model is an effective strategy to handle the nonlinear HSI data effectively. LPP-based SR (LLESR) and NPE-based SR (LESR) are two manifold-based SR algorithms that add the manifold regularization terms of LPP and NPE into the SR model [22]. Manifold regularized local SR (MRLSR) combines group SR model with local similarity regularization term to overcome illumination variations or occlusions problem in face recognition [23]. However, the manifold regularization term is unable to reveal the nonlinear manifold structure, and it is hard to select the optimal coefficient of the regularization term. Sparse manifold clustering and embedding (SMCE) introduces the local similarity distance into SR and comes into being a weighted sparse manifold representation model [24]. Then, a sparse discriminant manifold embedding (SDME) method is developed on the basis of SMCE to learn a projection for extracting embedding features [25]. SMCE only enforces the sparse coefficients between similar points to be large, but it cannot guarantee the sparse coefficients between dissimilar points to be zero. Therefore, these SR models are unable to consider the nonlinear characteristic and the local structure of HSI data.

In addition, the aforementioned SR-based DR methods generally conduct DR via a simple graph that only represents the one-to-one relationship of data [26]. In HSI, pixels often possess complex many-to-many relationships, such as one sample versus multiple neighborhoods or one neighborhood versus multiple samples [27], [28]. To reveal the multivariate relationships of data, some studies presented a hypergraph framework where the hyperedge is composed of multiple connected vertices [29]. Discriminant hyper-Laplacian projection (DHLP) constructs a hypergraph with label information to reveal the multivariate discriminant relation between samples [30]. Multistructure manifold embedding (MSME) designs an unsupervised sparse hypergraph and a geometrical graph to discover different intrinsic structures in high-dimensional data [31]. Despite the fact that these methods consider the multiple relationships between samples, they ignore the local sparse relation and the manifold structure in DR.

Through summarizing the abovementioned DR algorithms, we find there are two major challenges for DR of HSI. The first one is how to reveal the manifold structure and sparse relation of samples in high-dimensional nonlinear HSI data simultaneously. The other challenge is to preserve the multivariate sparse relationship and manifold structure of samples in the embedding feature space effectively. Inspired by these two motivations, we put forward a new supervised DR method termed local constraint-based sparse manifold hypergraph learning (LC-SMHL) for HSI classification. In LC-SMHL, a new SR model named local constrained sparse manifold coding (LCSMC) is built by combining manifold learning and SR. Then, two intraclass and interclass sparse manifold hypergraphs are designed to represent the intrinsic manifold structure and the multivariate sparse relationship between samples. The main contributions of this article include the following.

1) Compared with traditional SR methods, the proposed LC-SMHL designs a new manifold-based SR model LCSMC in which a local constraint has been imposed to guarantee that the sparse coefficients of nonneighborhood samples to be zero. Such local structure preservation makes the learned discriminant sparse graphs more effective for inheriting the local structure of original data.
2) Differing from the most manifold-based SR models that rely on manifold regularization term or reweight sparse coefficients, the LCSMC model explores sparse property via the local neighborhood on the manifold, and it explores the sparse relation between manifold neighborhoods. Therefore, LCSMC is able to discover the intrinsic manifold and sparse property of HSI via the manifold structure preservation in sparse optimization.
3) With hypergraph framework and LCSMC model, LC-SMHL constructs two within-class and between-class sparse hypergraphs to fully reveal the complex multivariate sparse relationship between samples, and the structure of hypergraphs is maintained to preserve the manifold structure and sparse relationship for enhancing the discriminant power of embedding data, which is helpful for subsequent HSI classification.

The rest of this article is organized as follows. Section II gives the details about the proposed DR method LC-SMHL and the designed SR model LCSMC. In Section III, some experiments are conducted on three real HSI data to demonstrate the effectiveness of the proposed method. Finally, Section IV summarizes this article and provides some suggestions for future work.

II. LC-SMHL

Based on the theories of manifold learning and SR, we propose a new discriminant DR method named LC-SMHL to reveal the manifold structure and the multivariate sparse relationship between samples. The flowchart of the proposed method is illustrated in Fig. 1.

For convenience, we first define an HSI data set $\mathbf{X} = [\mathbf{x}_1, \mathbf{x}_2, \ldots \mathbf{x}_n] \in \Re^{D\times n}$ with D spectral bands and n samples. The class label of $\mathbf{x}_i$ is denoted as $l_i \in \{1, 2, \ldots, c\}$, and c is the number of total classes. The corresponding embedding data are represented by $\mathbf{Y} = \left[\mathbf{y}_1, \mathbf{y}_2, \cdots \mathbf{y}_n\right] \in \Re^{d\times n}$ in which d is the number of embedding dimension. For linear DR methods, $\mathbf{Y}$ can be obtained by $\mathbf{Y} = \mathbf{V}^T\mathbf{X}$ with a projection matrix $\mathbf{V} \in \Re^{D\times d}$, where $d \ll D$.

A. LCSMC

Hyperspectral image (HSI) usually contains a complex nonlinear structure because the spectrum transmitting in the atmosphere will suffer from various nonlinear optical effects, such as reflection, absorption, and dispersion. Therefore, HSI data in high-dimension space are generally embedded in different manifolds. A universal hypothesis of manifold learning theory is that the nonlinear manifold in a small neighborhood

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 03:23:59 UTC from IEEE Xplore. Restrictions apply.

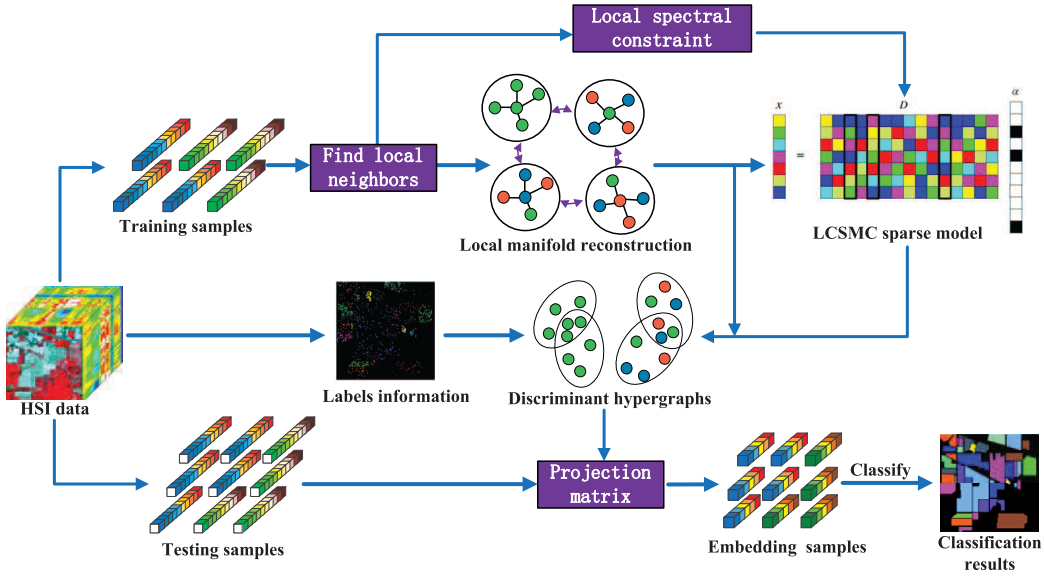


Fig. 1. Flowchart of the proposed method.

can be approximately regarded as a linear subspace. Under this hypothesis, the task of distinguishing different local nonlinear manifolds can be transformed into distinguishing the different linear subspaces in the local neighborhood. SR is an effective tool to reveal the structure of the subspace. Therefore, we introduce a local neighborhood constraint into a sparse model to design an LCSMC model for revealing the manifold structure and sparse relation of data.

For sample $\mathbf{x}_i$, m nearest neighbors (NNs) of $\mathbf{x}_i$ are selected according to the Euclidean distance. For neighbor $\mathbf{x}_{im}^{j_m}$, the subscript im denotes the mth neighbor of $\mathbf{x}_i$, while the superscript j_m indicates the sequence of sample $\mathbf{x}_{im}^{j_m}$ in the whole data set $\mathbf{X}$. The neighbor set $\Omega(\mathbf{x}_i)$ of $\mathbf{x}_i$ is constructed by

$$\Omega(\mathbf{x}_i) = \left\{\mathbf{x}_{i1}^{j_1}, \mathbf{x}_{i2}^{j_2}, \ldots, \mathbf{x}_{im}^{j_m}\right\}. \tag{1}$$

In order to preserve the manifold structure in data, $\mathbf{x}_i$ is geometrically reconstructed with neighbors in $\Omega(\mathbf{x}_i)$, and the reconstruction weights of $\mathbf{x}_i$ are calculated by the following objective function:

$$\min \sum_{i=1}^{n} \left\| \mathbf{x}_i - \sum_{t=1}^{m} e_{it}^{j_t} \mathbf{x}_{it}^{j_t} \right\|^2 \quad \text{s.t.} \sum_{t=1}^{m} e_{it}^{j_t} = 1 \tag{2}$$

where $e_{it}^{j_t}$ is the reconstruction weight between $\mathbf{x}_i$ and neighbor $\mathbf{x}_{it}^{j_t}$. With the weight constraint, the sum of reconstruction errors can be represented as

$$\begin{aligned} \left\| \mathbf{x}_i - \sum_{t=1}^{m} e_{it}^{j_t} \mathbf{x}_{it}^{j_t} \right\|^2 &= \left\| \sum_{t=1}^{m} e_{it}^{j_t} \left(\mathbf{x}_i - \mathbf{x}_{it}^{j_t} \right) \right\|^2 \\ &= \left\| \left[\mathbf{x}_i - \mathbf{x}_{i1}^{j_1}, \ldots, \mathbf{x}_i - \mathbf{x}_{it}^{j_t} \right] \mathbf{e}_i \right\|^2 = \mathbf{e}_i^T \mathbf{B}_i \mathbf{e}_i \end{aligned} \tag{3}$$

in which $\mathbf{e}_i = [e_{i1}^{j_1}, e_{i2}^{j_2}, \ldots, e_{im}^{j_m}]^T$ is local reconstruction weight vector, and symmetric matrix $\mathbf{B}_i$ equals to $\left[\mathbf{x}_i - \mathbf{x}_{i1}^{j_1}, \ldots, \mathbf{x}_i - \mathbf{x}_{it}^{j_t}\right]^T \left[\mathbf{x}_i - \mathbf{x}_{i1}^{j_1}, \ldots, \mathbf{x}_i - \mathbf{x}_{it}^{j_t}\right]$. Thus, the abovementioned objective function is reduced as

$$\arg\min_{\mathbf{e}_i} \ \mathbf{e}_i^T \mathbf{B}_i \mathbf{e}_i \quad \text{s.t.} \ \mathbf{i}^T \mathbf{e}_i = 1. \tag{4}$$

With the Lagrangian multipliers method, the optimization solution can be obtained as

$$\mathbf{e}_i = \frac{(\mathbf{B}_i)^{-1}\mathbf{i}}{\mathbf{i}^T (\mathbf{B}_i)^{-1}\mathbf{i}}. \tag{5}$$

For representing reconstruction point with the whole data set $\mathbf{X}$, we transform local reconstruction weight vector $\mathbf{e}_i$ into global reconstruction vector $\mathbf{g}_i = [g_{i1}, g_{i2}, \ldots, g_{in}]^T$ by

$$g_{ij} = \begin{cases} e_{iq}^{j_q}, & j = j_q \text{ and } q \in \{1, 2, \ldots, m\} \\ 0, & \text{otherwise} \end{cases} \tag{6}$$

and the reconstruction point of $\mathbf{x}_i$ can be represented as $\hat{\mathbf{x}}_i^g \approx \mathbf{X}\mathbf{g}_i$. The manifold reconstruction points represent the comprehensive manifold properties of the samples in local neighborhood, and the difference between two reconstruction points reflects the similarity degree of manifold structure in different local neighbor sets.

Relying on the vector $\mathbf{g}_i$, the manifold neighborhood error matrix $\mathbf{E}_i$ and the manifold similarity weight vector $\mathbf{P}_i$ of $\mathbf{x}_i$ are designed as (7) and (8). Matrix $\mathbf{E}_i$ represents the difference between sample $\mathbf{x}_i$ and other manifold reconstruction points, and its jth vector is computed by the normalized error between $\mathbf{x}_i$ and the reconstructed point $\mathbf{X}\mathbf{g}_i$ of sample $\mathbf{x}_j$. The manifold neighborhood error matrix $\mathbf{E}_i$ is defined by considering two aspects. At first, HSI data usually have complex manifold structure, and the simple error between $\mathbf{x}_i$ and $\mathbf{x}_j$ is insufficient to represent the intrinsic difference of two samples. Therefore, $\mathbf{E}_i$ calculates the error between $\mathbf{x}_i$ and the reconstruction point $\mathbf{X}\mathbf{g}_i$ to fully discover the difference between sample $\mathbf{x}_i$ and the manifold reconstruction point of $\mathbf{x}_j$. Furthermore, to eliminate scale impact of different sample, $\mathbf{E}_i$ utilizes the normalized

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 03:23:59 UTC from IEEE Xplore. Restrictions apply.

error between $\mathbf{x}_i$ and the reconstructed point $\mathbf{Xg}_i$ instead of exploiting error between $\mathbf{x}_i$ and $\mathbf{Xg}_i$ directly, which avoids the oversized samples bringing too larger influence for error term in LCSMC model.

For vector $\mathbf{P}_i$, its elements denote the spectral weights between sample $\mathbf{x}_i$ and reconstruction points. The reconstruction point possessing large spectral difference with $\mathbf{x}_i$ will have a large spectral weight, which will produce more inhibitory effect in sparse optimization and lead to a small sparse coefficient

$$\mathbf{E}_i = \left[\frac{\mathbf{x}_i - \mathbf{Xg}_1}{\|\mathbf{x}_i - \mathbf{Xg}_1\|}, \frac{\mathbf{x}_i - \mathbf{Xg}_2}{\|\mathbf{x}_i - \mathbf{Xg}_2\|}, \ldots, \frac{\mathbf{x}_i - \mathbf{Xg}_n}{\|\mathbf{x}_i - \mathbf{Xg}_n\|}\right] \in \Re^{D\times n} \tag{7}$$

$$\mathbf{P}_i = \left(\frac{\|\mathbf{x}_i - \mathbf{Xg}_1\|}{\sum_{t=1}^{n}\|\mathbf{x}_i - \mathbf{Xg}_t\|}, \frac{\|\mathbf{x}_i - \mathbf{Xg}_2\|}{\sum_{t=1}^{n}\|\mathbf{x}_i - \mathbf{Xg}_t\|}, \ldots, \frac{\|\mathbf{x}_i - \mathbf{Xg}_n\|}{\sum_{t=1}^{n}\|\mathbf{x}_i - \mathbf{Xg}_t\|}\right)^T \in \Re^{n\times 1}. \tag{8}$$

Based on the manifold neighborhood error matrix, manifold similarity weight vector, and local neighborhood, the objective function of the designed LCSMC model is constructed as follows:

$$\begin{cases} \min\limits_{\mathbf{s}_i} \lambda\|\mathbf{P}_i \otimes \mathbf{s}_i\|_0 + \dfrac{1}{2}\|\mathbf{E}_i\mathbf{z}_i\|_2^2 \\ \text{s.t. } \mathbf{s}_i = \mathbf{z}_i; \quad \mathbf{z}_i^T\mathbf{i} = 1; \ z_{ij} = 0, (i,j) \notin \Omega \end{cases} \tag{9}$$

where $\otimes$ is the dot product symbol, $\mathbf{s}_i$ is the sparse coefficient vector, $\mathbf{z}_i$ is a substitution variable of $\mathbf{s}_i$, and z_{ij} is the jth element of vector $\mathbf{z}_i$. λ is a tradeoff parameter balancing the sparsity and representation residual, Ω is a set of edges that define the neighborhood relationship between the samples in $\mathbf{X}$, and $(i, j) \notin \Omega$ indicates that $\mathbf{x}_i$ and $\mathbf{x}_j$ are not neighbors. In (9), the manifold similarity weight vector $\mathbf{P}_i$ is added into sparse item $\|\mathbf{P}_i \otimes \mathbf{s}_i\|_0$ to adjust the sparse coefficients according to the difference between samples and the local manifold reconstruction points of training samples, which promotes that the sample owning larger manifold reconstruction difference has a smaller sparse coefficient. Moreover, the local spectral constrain $z_{ij} = 0, (i, j) \notin \Omega$ is imposed to make the sparse coefficients between samples from different local spectral neighborhoods to be zero, which guarantees the sparse neighbors to be selected from the same manifold.

The visual sparse optimization process of LGSMC model is shown in Fig. 2. LCSMC seeks the sparse relations among manifold reconstruction points instead of original data points. Therefore, the obtained sparse coefficients are able to reflect the manifold similarity between samples and other manifold neighborhoods. In addition, by enforcing local spectral constrain $z_{ij} = 0, (i, j) \notin \Omega$, LCSMC meets the assumption that the nonlinear manifolds $\mathcal{M}_1$ and $\mathcal{M}_2$ in local neighborhood $\Omega(\mathbf{x}_1)$ can be approximately regarded as linear affine subspace $\mathcal{S}_1$ and $\mathcal{S}_2$. Therefore, LCSMC is able to select the sparse neighbors $\hat{\mathbf{x}}_2$ and $\hat{\mathbf{x}}_3$ from the same manifold and abandon neighbor $\hat{\mathbf{x}}_4$ that lies in other manifold.

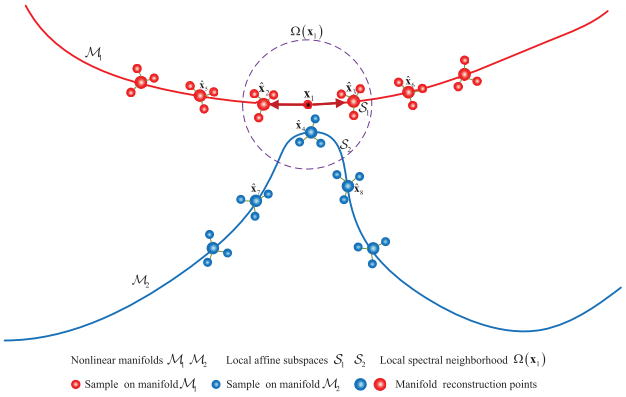


Fig. 2. Visualize sparse optimization of LCSMC.

B. Convex Optimization Solution of LCSMC

As the ℓ_0-norm in (9) is nonconvex function, we transform LCSMC into convex optimization model via convex relaxation techniques. By introducing a new vector $\mathbf{d}_i$ and impose an constraint $\mathbf{d}_i = \mathbf{z}_i$, the objective function of LCSMC can be written as

$$\begin{cases} \min\limits_{R_\Omega(\mathbf{d}_i)=\mathbf{0},\mathbf{s}_i,\mathbf{z}_i} \lambda\|\mathbf{P}_i \otimes \mathbf{s}_i\|_1 + \dfrac{1}{2}\|\mathbf{E}_i\mathbf{z}_i\|_2^2 \\ \text{s.t. } \mathbf{d}_i = \mathbf{s}_i; \quad \mathbf{d}_i = \mathbf{z}_i; \ \mathbf{d}_i^T\mathbf{i} = 1 \end{cases} \tag{10}$$

in which $R_{\bar{\Omega}}(\mathbf{z}_i)$ is a projection operator that makes elements z_{ij} of $\mathbf{z}_i$ equal to zero when $\mathbf{x}_j$ is not the local neighbor of $\mathbf{x}_i$, and $\mathbf{0}$ is an all-zero vector.

According to the Lagrange multiplier method, the augmented Lagrangian function (10) without the constraint $R_\Omega(\mathbf{z}_i) = \mathbf{0}$ is given by

$$\begin{aligned} &L(\mathbf{z}_i, \mathbf{s}_i, \mathbf{d}_i, \mathbf{T}_1, \mathbf{T}_2, T_3) \\ &= \lambda\|\mathbf{P}_i\mathbf{s}_i\|_1 + \frac{1}{2}\|\mathbf{E}_i\mathbf{z}_i\|_2^2 \\ &\quad + \langle\mathbf{T}_1, \mathbf{d}_i - \mathbf{z}_i\rangle + \langle\mathbf{T}_2, \mathbf{d}_i - \mathbf{s}_i\rangle + \langle T_3, 1 - \mathbf{i}^T\mathbf{d}_i\rangle \\ &\quad + \frac{\rho}{2}\left(\|\mathbf{d}_i - \mathbf{z}_i\|_F^2 + \|\mathbf{d}_i - \mathbf{s}_i\|_F^2 + \left\|1 - \mathbf{i}^T\mathbf{d}_i\right\|_F^2\right) \end{aligned} \tag{11}$$

in which vectors $\mathbf{T}_1$, $\mathbf{T}_2$, and scalar T_3 are the Lagrange multipliers corresponding to three equality constraints, respectively. According to the alternating direction method of multipliers (ADMMs), the variables $\mathbf{s}_i$, $\mathbf{z}_i$, and $\mathbf{d}_i$ are updated by minimizing $\mathcal{L}(\mathbf{z}_i, \mathbf{s}_i, \mathbf{d}_i, \mathbf{T}_1, \mathbf{T}_2, T_3)$ alternately.

For $\mathbf{z}_i$ and $\mathbf{s}_i$, they both have a closed-form proximal function solution by assuming other variables are fixed, and $\mathbf{z}_i^{k+1}$ and $\mathbf{s}_i^{k+1}$ in the kth iteration can be calculated by

$$\begin{aligned} \mathbf{z}_i^{k+1} &= \arg\min_{\mathbf{z}_i} \frac{1}{2}\|\mathbf{E}_i\mathbf{z}_i\|_2^2 + \frac{\rho}{2}\left\|\mathbf{d}_i - \mathbf{z}_i + \frac{\mathbf{T}_1^k}{\rho}\right\|_F^2 \\ &= \left(\mathbf{E}_i^T\mathbf{E}_i + \rho\mathbf{i}\right)^{-1}\left(\rho\mathbf{d}_i^k + \mathbf{T}_1^k\right) \end{aligned} \tag{12}$$

$$\begin{aligned} \mathbf{s}_i^{k+1} &= \arg\min_{\mathbf{s}_i} \lambda\|\mathbf{P}_i\mathbf{s}_i\|_1 + \frac{\rho}{2}\left\|\mathbf{d}_i^k - \mathbf{s}_i + \frac{\mathbf{T}_2^k}{\rho}\right\|_F^2 \\ &= \mathcal{S}_{\frac{\lambda\mathbf{P}_i}{\rho}}\left(\mathbf{d}_i^k + \frac{\mathbf{T}_2^k}{\rho}\right) \end{aligned} \tag{13}$$

in which $\mathcal{S}$ is the soft-thresholding operator defined as $\mathcal{S}_a(b) = \text{sgn}(b)(|b| - a)_+$.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 03:23:59 UTC from IEEE Xplore. Restrictions apply.

For variable $\mathbf{d}_i$, we derive a new update rule because it is restricted by nonlinear constraints $R_{\bar{\Omega}}(\mathbf{d}_i) = \mathbf{0}$. According to (10), $\mathbf{d}_i^{k+1}$ can be contained by solving the following problem:

$$\arg\min_{R_{\bar{\Omega}}(\mathbf{d}_i)=\mathbf{0}} \left\langle \left(\mathbf{d}_i^k - \mathbf{z}_i^{k+1} + \frac{\mathbf{T}_1^k}{\rho}\right) + \left(\mathbf{d}_i^k - \mathbf{s}_i^{k+1} + \frac{\mathbf{T}_2^k}{\rho}\right) - \mathbf{i}\left(1 - \mathbf{i}^T\mathbf{d}_i^k + \frac{T_3^k}{\rho}\right), \mathbf{d}_i - \mathbf{d}_i^k \right\rangle + \frac{\eta}{2}\left\|\mathbf{d}_i - \mathbf{d}_i^k\right\| \tag{14}$$

in which $\eta = 2 + \left\|\mathbf{i}^T\right\|_F^2$. Then, we introduce a new vector $\mathbf{q}_i^k$ and make it equal to

$$\mathbf{q}_i^k = \left(\mathbf{d}_i^k - \mathbf{z}_i^{k+1} + \frac{\mathbf{T}_1^k}{\rho}\right) + \left(\mathbf{d}_i^k - \mathbf{s}_i^{k+1} + \frac{\mathbf{T}_2^k}{\rho}\right) - \mathbf{i}\left(1 - \mathbf{i}^T\mathbf{d}_i^k + \frac{T_3^k}{\rho}\right). \tag{15}$$

Finally, the value of $\mathbf{d}_i^{k+1}$ can be computed by

$$\begin{aligned}\mathbf{d}_i^{k+1} &= \arg\min_{R_{\bar{\Omega}}(\mathbf{d}_i)=\mathbf{0}} \left\langle \mathbf{q}_i^k, \mathbf{d}_i - \mathbf{d}_i^k \right\rangle + \frac{\eta}{2}\left\|\mathbf{d}_i - \mathbf{d}_i^k\right\| \\ &= \arg\min_{R_{\bar{\Omega}}(\mathbf{d}_i)=\mathbf{0}} \frac{\eta}{2}\left\|\mathbf{d}_i - \mathbf{d}_i^k + \frac{\mathbf{q}_i^k}{\eta}\right\| = R_{\bar{\Omega}}\left(\mathbf{d}_i^k - \frac{\mathbf{q}^k}{\eta}\right).\end{aligned} \tag{16}$$

After obtaining $\mathbf{z}_i^{k+1}$, $\mathbf{s}_i^{k+1}$, and $\mathbf{d}_i^{k+1}$, the Lagrange multipliers $\mathbf{T}_1^{k+1}$, $\mathbf{T}_2^{k+1}$, and T_3^{k+1} can be updated with simple dual-ascend steps

$$\mathbf{T}_1^{k+1} = \mathbf{T}_1^k + \rho\left(\mathbf{d}_i^{k+1} - \mathbf{z}_i^{k+1}\right) \tag{17}$$

$$\mathbf{T}_2^{k+1} = \mathbf{T}_2^k + \rho\left(\mathbf{d}_i^{k+1} - \mathbf{s}_i^{k+1}\right) \tag{18}$$

$$T_3^{k+1} = T_3^k + \rho\left(1 - \mathbf{i}^T\mathbf{d}_i^{k+1}\right). \tag{19}$$

C. Sparse Manifold Hypergraph Learning

To reveal the multivariate relationships between samples of HSI, a sparse manifold hypergraph learning (LC-SMHL) method is proposed under hypergraph framework. With label information and the obtained sparse coefficients, a within-class hypergraph $\mathbf{G}^w = \{\mathbf{X}, \mathbf{E}^w, \mathbf{W}^w\}$ and a between-class hypergraph $\mathbf{G}^b = \{\mathbf{X}, \mathbf{E}^b, \mathbf{W}^b\}$ are constructed to represent the multivariate discriminant relationship between samples. $\mathbf{X}$ is the vertex set, and $\mathbf{E}^w$ and $\mathbf{E}^b$ are hyperedge sets, while $\mathbf{W}^w$ and $\mathbf{W}^b$ are weights matrices of $\mathbf{G}^w$ and $\mathbf{G}^b$, respectively.

For within-class hypergraph $\mathbf{G}^w$, the weight of hyperedge e_i^w is calculated by

$$w\left(e_i^w\right) = \sum_{\mathbf{x}_j \in e_i^w} s_{ij} \exp\left(-\frac{K_i^w\left\|\mathbf{x}_i - \mathbf{X}\mathbf{g}_j\right\|}{\sum_{t=1}^{K_i^w}\left\|\mathbf{x}_i - \mathbf{X}\mathbf{g}_t\right\|}\right) \tag{20}$$

where K_i^w is the number of intraclass sparse neighbors of $\mathbf{x}_i$. The within-class weight matrix $\mathbf{W}^w$ is defined as $\mathbf{W}^w = \operatorname{diag}\left(\left[w\left(e_1^w\right), \cdots, w\left(e_n^w\right)\right]\right)$. In (20), the weight is calculated based on the sparse coefficient s_{ij} and manifold reconstruction vector g_j. Thus, the hyperedge weight $w\left(e_i^w\right)$ is able to reflect the sparse relation and manifold structure of data, simultaneously.

To defined the relationship between each vertex and each hyperedge in $\mathbf{G}^w$, a point-edge correlation matrix $\mathbf{H}^w = \left[h\left(\mathbf{x}_i, e_j^w\right)\right]_{i,j}$ is defined as

$$h\left(\mathbf{x}_i, e_j^w\right) = \begin{cases} 1, & \text{if } s_{ji} \neq 0 \text{ and } l_i = l_j \\ 0, & \text{otherwise.} \end{cases} \tag{21}$$

With the within-class matrix weights $\mathbf{W}^w$ and incidence matrix $\mathbf{H}^w$, the degrees of each hyperedge e_i^w and vertex $\mathbf{x}_i$ in $\mathbf{G}^w$ can be obtained as follows:

$$d\left(e_j^w\right) = \sum_{i=1}^n h\left(\mathbf{x}_i, e_j^w\right), d(\mathbf{x}_i|w) = \sum_{j=1}^n h\left(\mathbf{x}_i, e_j^w\right) w\left(e_j^w\right). \tag{22}$$

Similarly, the hyperedge weight matrix $\mathbf{W}^b$ and incidence matrix $\mathbf{H}^b$ of between-class hypergraph $\mathbf{G}^b$ are defined as

$$w\left(e_i^b\right) = \sum_{\mathbf{x}_j \in e_i^b} s_{ij} \exp\left(-\frac{K_i^b\left\|\mathbf{x}_i - \mathbf{X}\mathbf{g}_j\right\|}{\sum_{t=1}^{K_i^b}\left\|\mathbf{x}_i - \mathbf{X}\mathbf{g}_t\right\|}\right) \tag{23}$$

$$h\left(\mathbf{x}_i, e_j^b\right) = \begin{cases} 1, & \text{if } s_{ji} \neq 0 \text{ and } l_i \neq l_j \\ 0, & \text{otherwise.} \end{cases} \tag{24}$$

The corresponding degrees of each hyperedge e_i^b and vertex $\mathbf{x}_i$ in $\mathbf{G}^b$ are obtained by

$$d\left(e_j^b\right) = \sum_{i=1}^n h\left(\mathbf{x}_i, e_j^b\right), d(\mathbf{x}_i|b) = \sum_{j=1}^n h\left(\mathbf{x}_i, e_j^b\right) w\left(e_j^b\right). \tag{25}$$

Based on the abovementioned two discriminant sparse hypergraphs, the objective function of LC-SMHL is constructed as (26) and (27) in which $\mathbf{L}^w$ is the within-class hyper-Laplacian matrix, and $\mathbf{L}^b$ is the between-class hyper-Laplacian matrix. $\mathbf{D}_x^w$ and $\mathbf{D}_e^w$ are the diagonal matrices of within-class vertex degree $d(\mathbf{x}_i|w)$ and hyperedge degree $d\left(e_j^w\right)$. $\mathbf{D}_x^b$ and $\mathbf{D}_e^b$ are the diagonal matrices of within-class vertex degree $d(\mathbf{x}_i|b)$ and hyperedge degree $d\left(e_j^b\right)$, respectively

$$\begin{aligned}&\min_{\mathbf{V}} \frac{1}{2}\sum_{e_k^b \in E^b} \frac{w\left(e_k^b\right)h\left(\mathbf{x}_i, e_k^b\right)h\left(\mathbf{x}_j, e_k^b\right)}{d\left(e_k^b\right)} \times \sum_{\mathbf{x}_i, \mathbf{x}_j \in \mathbf{X}} \left\| \frac{\mathbf{V}^T\mathbf{x}_i}{\sqrt{d(\mathbf{x}_i|b)}} - \frac{\mathbf{V}^T\mathbf{x}_j}{\sqrt{d\left(\mathbf{x}_j|b\right)}} \right\|_2^2 \\ &= \min_{\mathbf{V}} tr\left\{\mathbf{V}^T\mathbf{X}\left[\mathbf{D}_x^b - \mathbf{H}^b\mathbf{W}^w\left(\mathbf{D}_e^b\right)^{-1}\left(\mathbf{H}^b\right)^T\right]\mathbf{X}^T\mathbf{V}\right\} \\ &= \min_{\mathbf{V}} tr\left\{\mathbf{V}^T\mathbf{X}\mathbf{L}^b\mathbf{X}^T\mathbf{V}\right\}\end{aligned} \tag{26}$$

$$\begin{aligned}&\min_{\mathbf{V}} \frac{1}{2}\sum_{e_k^w \in E^w} \frac{w\left(e_k^w\right)h\left(\mathbf{x}_i, e_k^w\right)h\left(\mathbf{x}_j, e_k^w\right)}{d\left(e_k^w\right)} \times \sum_{\mathbf{x}_i, \mathbf{x}_j \in \mathbf{X}} \left\| \frac{\mathbf{V}^T\mathbf{x}_i}{\sqrt{d(\mathbf{x}_i|w)}} - \frac{\mathbf{V}^T\mathbf{x}_j}{\sqrt{d\left(\mathbf{x}_j|w\right)}} \right\|_2^2 \\ &= \min_{\mathbf{V}} tr\left\{\mathbf{V}^T\mathbf{X}\left[\mathbf{D}_x^w - \mathbf{H}^w\mathbf{W}^w\left(\mathbf{D}_e^w\right)^{-1}\left(\mathbf{H}^w\right)^T\right]\mathbf{X}^T\mathbf{V}\right\} \\ &= \min_{\mathbf{V}} tr\left\{\mathbf{V}^T\mathbf{X}\mathbf{L}^w\mathbf{X}^T\mathbf{V}\right\}.\end{aligned} \tag{27}$$

According to this pair of objective functions, the intraclass samples should be compacted as close as possible, while the

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 03:23:59 UTC from IEEE Xplore. Restrictions apply.

interclass samples should be separated as far as possible in the embedding space, and the optimal projection $\mathbf{V}$ can be learned by the following objective function:

$$\begin{cases} \min\limits_{\mathbf{V}} \mathbf{V}^T\mathbf{X}\left(\alpha\mathbf{L}^w - \mathbf{L}^b\right)\mathbf{X}^T\mathbf{V} \\ \text{s.t. } \mathbf{V}^T\mathbf{X}\left(\alpha\mathbf{D}_x^w - \mathbf{D}_x^b\right)\mathbf{X}^T\mathbf{V} = \mathbf{I} \end{cases} \tag{28}$$

where α is a nonnegative tradeoff parameter to balance the effectiveness of intraclass and interclass information in optimization. The constraint in objective function is constructed to avoid the trivial solution. If without the constraint, the zero matrix will be the optimal solution for $\mathbf{V}$, but it is meaningless for dimensional reduction. With the Lagrangian multiplier method, the solution of (28) is transformed as the Lagrangian function

$$\mathcal{L}(\mathbf{V},\beta) = \mathbf{V}^T\mathbf{X}\left(\alpha\mathbf{L}^w - \mathbf{L}^b\right)\mathbf{X}^T\mathbf{V} + \beta\left[\mathbf{I} - \mathbf{V}^T\mathbf{X}\left(\alpha\mathbf{D}_x^w - \mathbf{D}_x^b\right)\mathbf{X}^T\mathbf{V}\right] \tag{29}$$

in which β is the multiplier. By setting the derivative of the Lagrangian function as zero, the optimal solution of $\mathbf{V}$ can be transformed to solve the following generalized eigenvalue problem:

$$\mathbf{X}\left(\alpha\mathbf{L}^w - \mathbf{L}^b\right)\mathbf{X}^T\mathbf{v}_i = \beta_i\mathbf{X}\left(\alpha\mathbf{D}_x^w - \mathbf{D}_x^b\right)\mathbf{X}^T\mathbf{v}_i \tag{30}$$

where $\mathbf{v}_i$ is a eigenvector and β_i denotes corresponding eigenvalue. With the d smallest eigenvalues corresponding eigenvectors, the optimal projection can be denoted as

$$\mathbf{V} = [\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_d] \in \Re^{D\times d} \tag{31}$$

and the embedding data can be represented by $\mathbf{Y} = \mathbf{V}^T\mathbf{X} \in \Re^{d\times n}$. After complicated theoretical derivation processing, the optimal projection matrix only needs to solve the above-generalized eigenvalue problem. Therefore, the actual program operation process of the proposed method will not consume huge computing resources and running time, which is beneficial to expand its practical application.

According to the abovementioned formulas, we utilize the big O notation to analyze the computational complexity of the proposed method. The number of training samples is n_0, and the maximum number of sparse coding iterations is t. m and D denote the number of local neighbors and spectral bands, respectively. The process of performing local manifold reconstruction takes $O\left(n_0m^2\right)$. The manifold neighborhood error matrix $\mathbf{E}_i$ and the manifold similarity weight vector $\mathbf{P}_i$ of all training samples are computed with the cost of $O\left(2n_0^2\right)$. The cost of LCSMC is $O\left(tn_0^2\right)$. The construction of sparse manifold hypergraphs $\mathbf{G}^w$ and $\mathbf{G}^b$ both spend $O\left(2n_0^2\right)$, while the calculation of hyper-Laplacian matrices $\mathbf{L}^w$ and $\mathbf{L}^b$ both cost $O\left(n_0^3\right)$. It takes $O\left(D^3\right)$ to learn the optimal projection matrix via the final objective function. For $m < n_0$, the total computational complexity of LC-SMHL is $O\left(n_0^3 + tn_0^2 + D^3\right)$ that mainly depends on the number of training samples, iterations, and spectral bands.

III. Experimental Results and Analysis

In this section, we employed three popular HSI data sets to validate the proposed LC-SMHL method and LCSMC model and compared them with some state-of-art DR methods and SR models to demonstrate their merits.

A. Data Sets

1) Indian Pines Data Set: It contains 145×145 pixels and 220 spectral bands that cover the spectral range from 375 to 2500 nm in 1992. In experiments, the 104–108,150–163, and 220 spectral bands are removed to avoid the noise and water absorption phenomena. A total of 16 classes of ground objects are considered in the data set. Its ground-truth map and class list are shown in Fig. 3(a).

2) Heihe Data Set: This image was acquired by the Compact Airborne Spectrographic Imager (CASI) sensor over Heihe watershed, Gansu Province, China, in 2015. It consists of 684×453 pixels, and 103 bands were used for experiments after removing water-absorption bands [37]. There are eight classes of ground objects considered in this scene. The ground truth and class list of HSI data are exhibited in Fig. 3(b).

3) Salinas Data Set: It was also collected by the AVIRIS sensor over the southern region of California, and its size is up to 512×217 pixels. This HSI captures the spectrum from 400 to 2500 nm and records the spectral values in 224 bands. Sixteen land cover types are considered in the experiment. The ground-truth map and class list are given in Fig. 3(c).

B. Experimental Setup

To verify the merits of the proposed DR method, we compared LC-SMHL with several state-of-art DR methods, including one statistics-based method principal component analysis (PCA) [32], three manifold-based methods LPP [8], NPE [9] and LGSFA [11], five representation-based methods SPP [16], DLSP [17], SLGDA [18], SDME [25], low-rank graph-based discriminant analysis (LGDA) [33], and two hypergraph-based methods DHLP [30] and MSME [31]. In addition, two deep learning methods—deep belief net (DBN) and stacked convolution sparse autoendoer (SAE)—were also exploited for comparison [38]. The raw spectral feature (RSF) of hyperspectral was also used in experiments.

The optimized parameters of the DR algorithm are obtained via cross-validation. For NPE and LPP, nine within-class neighbors are selected for each sample. In LGSFA and DHLP, the numbers of intraclass neighbors and interclass neighbors are set to 10 and 30, respectively. The sparse degrees were optimistically set to 10 for SPP, MSME, and SLGDA, while the regularization parameters in DLSP, LGDA, and SDME are, respectively, set to 0.5, 0.1, and 0.01. For DBN and SAE, we used the DeepLearnToolbox in a MATLAB version.

After obtaining embedding features with different DR methods, NN classifier, support vector machine (SVM), and SVM based on composite kernels (SVMCK) [34] were exploited for classification. Overall classification accuracy (OA), average classification accuracy (AA), and kappa coefficient (KC) were employed to evaluate the classification results. All experiments were performed on a personal computer with the i5-4590 central processing unit, 8-GB memory, and 64-bit Windows 10 using MATLAB 2014a.

C. Parameters Tuning

The proposed LC-SMHL method possesses three parameters including the tradeoff parameter α, the sparse

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 03:23:59 UTC from IEEE Xplore. Restrictions apply.

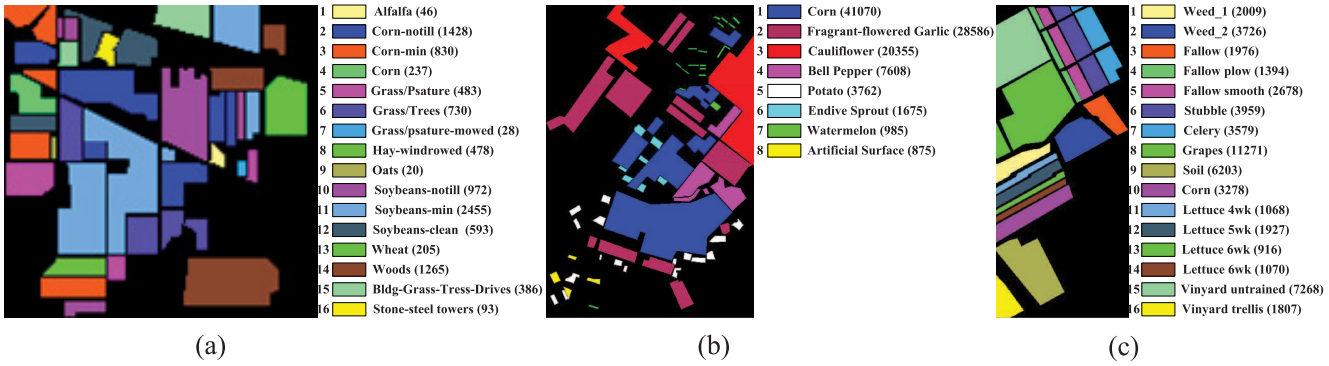


Fig. 3. Ground-truth maps and classes lists of three HSI data sets. (a) Indian Pines. (b) Heihe. (c) Salinas.

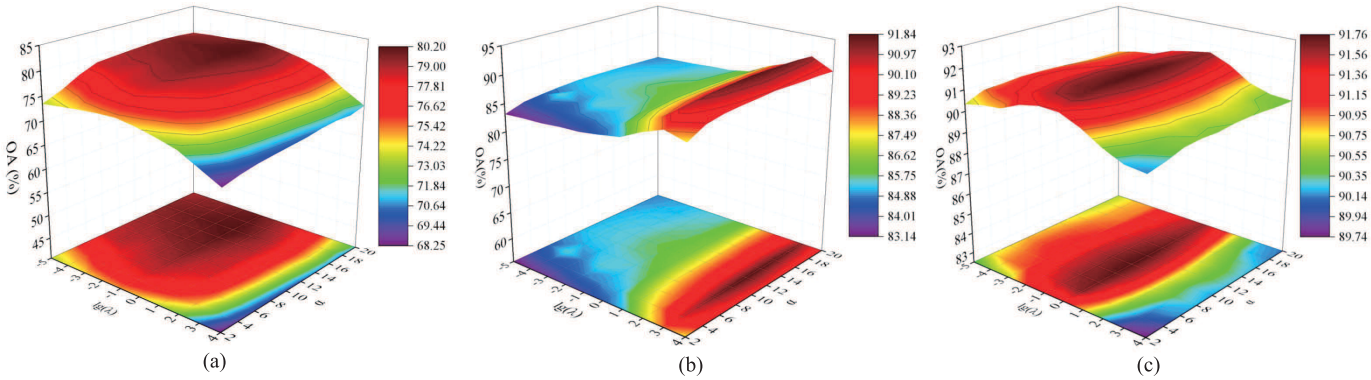


Fig. 4. OA versus different parameters α and λ. (a) Indian Pines. (b) Heihe. (c) Salinas.

regularization parameter λ, and the number of local neighbors m. To achieve the optimal embedding model, we performed parameter tuning experiments on three HSI data sets. In the Indian Pines data set, 10% samples from each class were selected as the training set, while 2% samples per class in Salinas and Heihe data sets were selected as the training sets. NN classifier was used to classify the remaining test samples. Each experiment was repeated ten times to reduce random errors. The average OAs with respect to λ and α are exhibited in Fig. 4, while the relationship between OAs and m is shown in Fig. 5.

According to Fig. 4, the parameter λ brings a larger influence on the classification accuracies of three data sets, which indicates that the balance between sparsity and representation residual is important for LCSMC model to reveal the sparse relation between samples. With the increase in λ, the OAs first rise and then turn to decline, while the change in α has smaller impacts on three data sets compared with λ. To achieve optimal parameters, we set λ and α to 10^{-1} and 16 for the Indian Pines data set, 10^3 and 12 for the Heihe data set, and 10^2 and 10 for the Salinas data set. In Fig. 5, the OAs are improved with the growth of m since a large local neighborhood is conducive to preserve the local structure adequately. When m keeps increasing, the OAs decrease on the Indian Pines data set and tend to slightly decline on the Heihe and Salinas data sets because an oversize local neighborhood cannot guarantee all neighbors belong to the same class. According to the OAs curves in Fig. 5, we set m as an optimal value of 80 for the

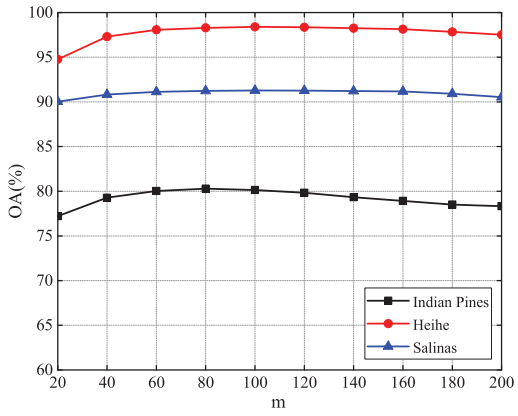


Fig. 5. OA with respect to different numbers of neighbors m.

Indian Pines data set and 100 for the Heihe and Salinas data sets.

When we meet unknown HSI data sets in the real application, we will select the optimal parameters with two approaches. First, we will randomly select a small part of the samples and manually calibrate their labels. The calibrated sample set includes the basic information of the unknown HSI data set, and it can be used as the training set for the supervised learning algorithm. By the means of the calibrated sample set, we also can perform the experiment of parameter turning to select the optimal parameters for unknown HSI data sets. In addition, if calibrated samples cannot be obtained in

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 03:23:59 UTC from IEEE Xplore. Restrictions apply.

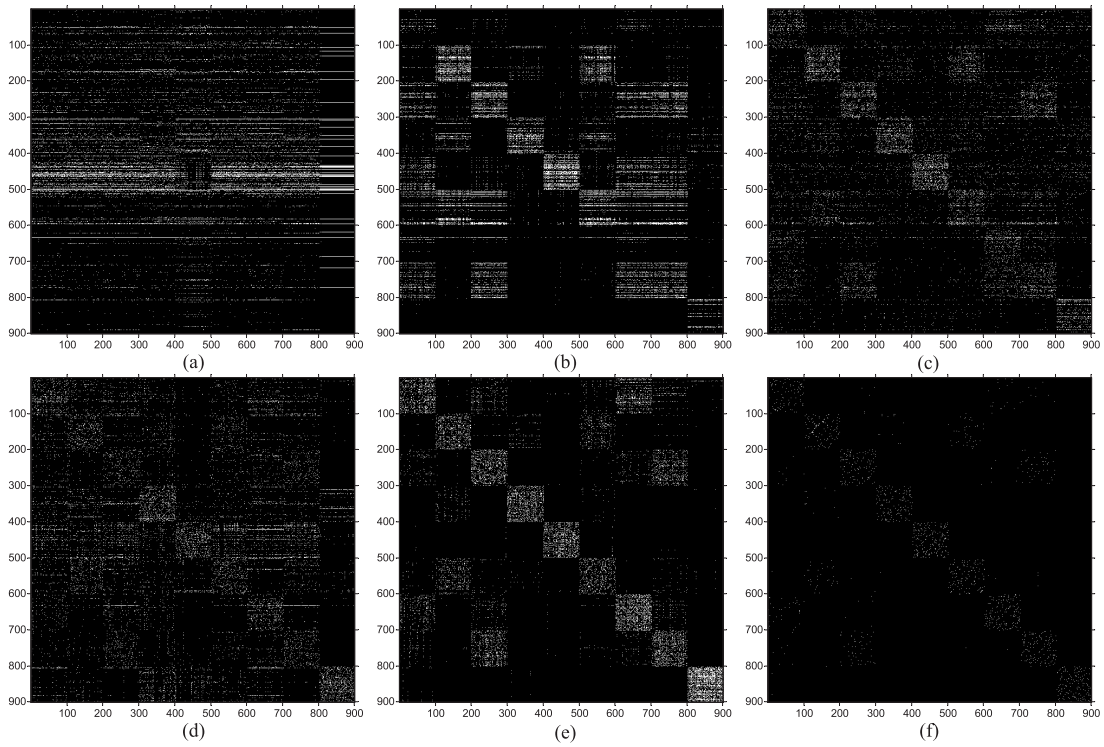


Fig. 6. Visualized sparse coefficients matrices obtained by different SR models. (a) SOMP. (b) BP. (c) SLRR. (d) MSRR. (e) SMCE. (f) LCSMC.

real practice, we will determine the parameters via experience learned from the parameter experiments on the Indian Pines, Salinas, and Heihe data sets. According to the parameter turning experimental results on the abovementioned three data sets, we can set $m = 100$, $\alpha = 12$, and $\lambda = 10^1$ for an unknown HSI data set. The optimal parameters obtained by experience may not be very precise, but they can provide a suitable range of appropriate values.

D. Analysis for LCSMC Model

The designed LCSMC model is a key step for LC-SMHL to capture the sparse neighbors and reveal the manifold similarity between samples. To explore the performance of LCSMC, we compare it with some SR models, such as simultaneous orthogonal matching pursuit (SOMP) [35], basis pursuit (BP) [36], sparse and low-rank representation (SLRR), manifold-constrained SR (MSRR), and SMCE. The first model belongs to greedy algorithms, while the latter four models are convex optimization algorithms.

To analyze the difference between these SR models, we randomly selected 900 samples from the Indian Pines data set to construct a nine-class HSI data set (100 samples per class). These SR models were conducted on this HSI data set. The visualized sparse coefficients' matrices and SR residuals of different SR models are shown in Figs. 6 and 7, respectively.

As shown in Fig. 6(a) and (d), SOMP and SMRR do not exhibit a block-diagonal structure for revealing high interclass separability. This means that the sparse coefficient matrices of them are not suitable for discriminant DR or supervised classification. In Fig. 6(e) and (f), the coefficients matrices obtained by SMCE and LCSMC represent the most obvious block-diagonal structure since they both consider the nonlinear manifold structure during the optimization process. In comparison with SMCE, the designed LCSMC model produces the sparsest coefficients matrix because its local spectral constraint enforces the coefficients between faraway points to be zero and guarantees all sparse neighbors selected from the same manifold. Therefore, the lower sparsity and higher between-class separability of LCSMC are suitable for constructing more effective sparse graphs.

For SR residuals shown in Fig. 7, we can evaluate SR models according to the average representation residuals (ARRs) and the standard deviation (STD) of representation residuals. SOMP and SLRR in Fig. 7(a) and (c) achieve the smallest ARR and STD at the expense of a sparse degree, which leads to their sparse coefficients to not reveal the sparse relationship between data points. Among all manifold-based SR model, the designed LCSMC model possesses the optimal ARR and STD compared with MSRR and SMCE. These experimental results indicate that LCSMC is able to accurately represent each sample with as few sparse neighbors as possible for achieving good sparsity degree and representation residual simultaneously.

In order to explore the impact of SR models on DR, we replaced the LCSMC in LC-SMHL with the aforementioned five sparse models to obtain five modified DR algorithms; 10% samples of the Indian Pines data set and 2% samples of the Heihe and Salinas data sets were, respectively, selected for training, while the remained samples were tested by NN and SVM. Experiments were repeated ten times in each condition. The OAs with STD (in percent) and average computing times (ACTs) (in second) of different algorithms are shown in Table I. ACT only includes the running time of DR but did not include the running time of classification.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 03:23:59 UTC from IEEE Xplore. Restrictions apply.

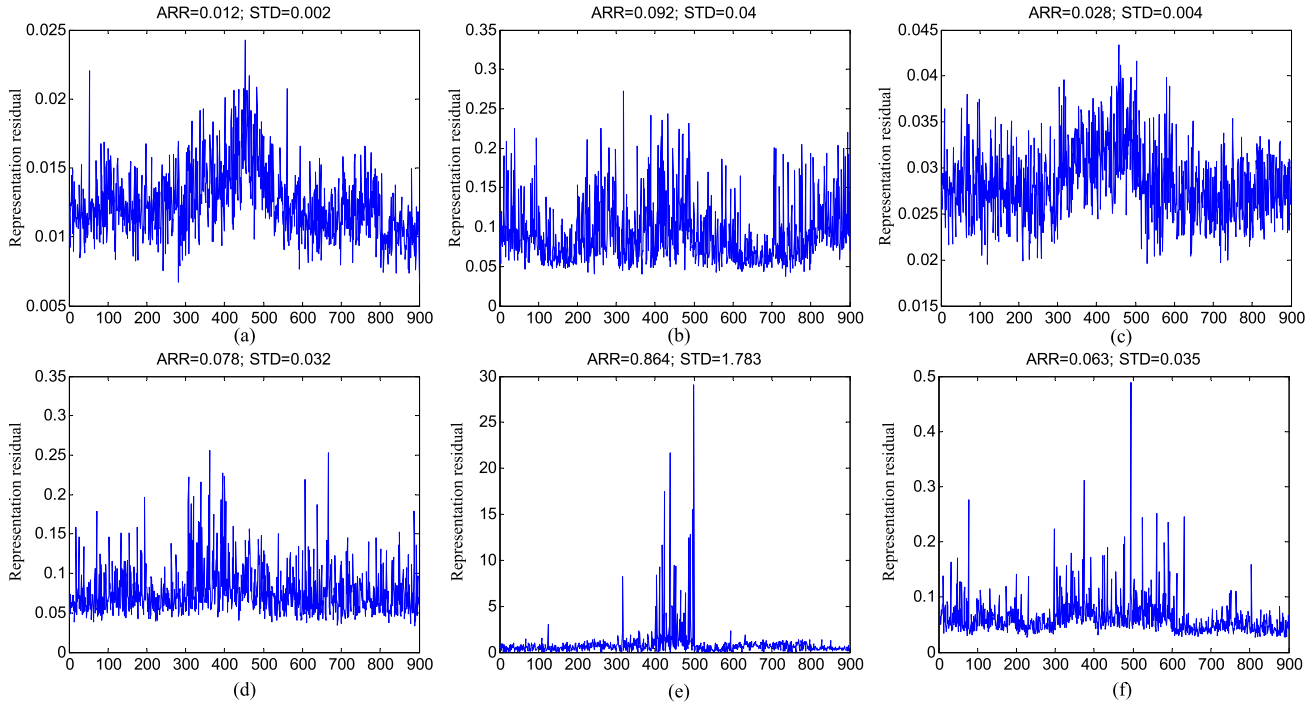


Fig. 7. Residuals of samples sparse represented by different SR models. (a) SOMP. (b) BP. (c) SLRR. (d) MSRR. (e) SMCE. (f) LCSMC.

TABLE I

CLASSIFICATION RESULTS OF LC-SMHL COMBINED WITH DIFFERENT SR MODELS. [OA ± STD (%), ACT (S)]

Datasets	Indian Pines			Heihe			Salinas		
Models	NN	SVM	ACT	NN	SVM	ACT	NN	SVM	ACT
SOMP	69.21±1.64	74.54±0.61	659.8	93.04±0.65	94.57±0.58	689.3	87.21±0.16	89.35±0.56	251.3
BP	73.04±1.41	76.71±0.72	36.8	94.59±0.66	95.04±0.39	74.2	89.83±0.29	91.72±0.23	16.1
SLRR	73.71±0.96	76.65±1.28	154.9	94.88±0.63	95.24±0.57	266.3	89.53±0.26	91.07±0.12	41.5
MSRR	71.62±1.29	75.68±1.48	304.9	93.69±0.47	93.14±0.85	1528.9	88.18±0.22	90.35±0.63	186.3
SMCE	76.80±1.21	78.82±1.47	368.8	95.83±0.46	96.12±0.64	1589.6	90.54±0.08	91.85±0.23	152.7
LCSMC	79.66±0.78	80.44±0.91	132.9	97.81±0.53	97.45±0.55	1126.9	91.82±0.43	92.43±0.37	86.3

Note: ACT only represents the running time of dimensionality reduction.

LC-SMHL combining with SMCE and LCSMC achieves better classification results in comparison with other methods due to they both consider the manifold structure in sparse optimization. LCSMC achieves the best results among all SR models no matter combined with NN or SVM because it introduces manifold neighborhood reconstruction and local spectral constraints into the sparse model. Although LCSMC has a complex mathematical theory, it costs the least running time among all compared SR models except the BP model. The reason is that LCSMC only conducts the parse optimization process in local neighborhoods to further reduce the computational complexity. In addition, the computational complexity of LCSMC is mainly depended on the number of training samples. In real applications, we usually utilize small samples set for training due to the limitation of labeled data, so the LCSMC model will not spend too much time.

According to the abovementioned experimental results and analysis, it is worthwhile to highlight several merits of LCSMC over the existing SR models.

1) Benefiting from the manifold similarity weights, LCSMC is able to accurately represent each sample with as fewer sparse neighbors as possible for achieving good sparsity degree and representation residual simultaneously.
2) Based on the local spectral constraint, the LCSMC model enforces the coefficients between faraway points to be zero and guarantees all sparse neighbors selected from the same manifold.
3) LCSMC optimally selects sparse neighbors in the local neighborhood instead of the whole training set, which reduces the computational complexity and running time significantly.

E. Investigate of Embedding Performance

To visually illustrate the effectiveness of LC-SMHL, we construct a three-class synthetic data set by selecting some samples from three classes of the Indian pines data set (green, red, and blue points represent the samples from classes Corn, Corn-min, and Corn-notill, respectively). LC-SMHL and some existing DR methods map this synthetic data set into 3-D space, and the distributions of embedding data are shown in Fig. 8.

From Fig. 8, it is clearly seen that these three classes of points usually overlap because three land cover types possess similar spectral features. In Fig. 8(a), (e), and (j), the embedding data points obtained by PCA, SPP, and MSME produce serious overlap phenomenon, because these three unsupervised methods ignore the discriminant information

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 03:23:59 UTC from IEEE Xplore. Restrictions apply.

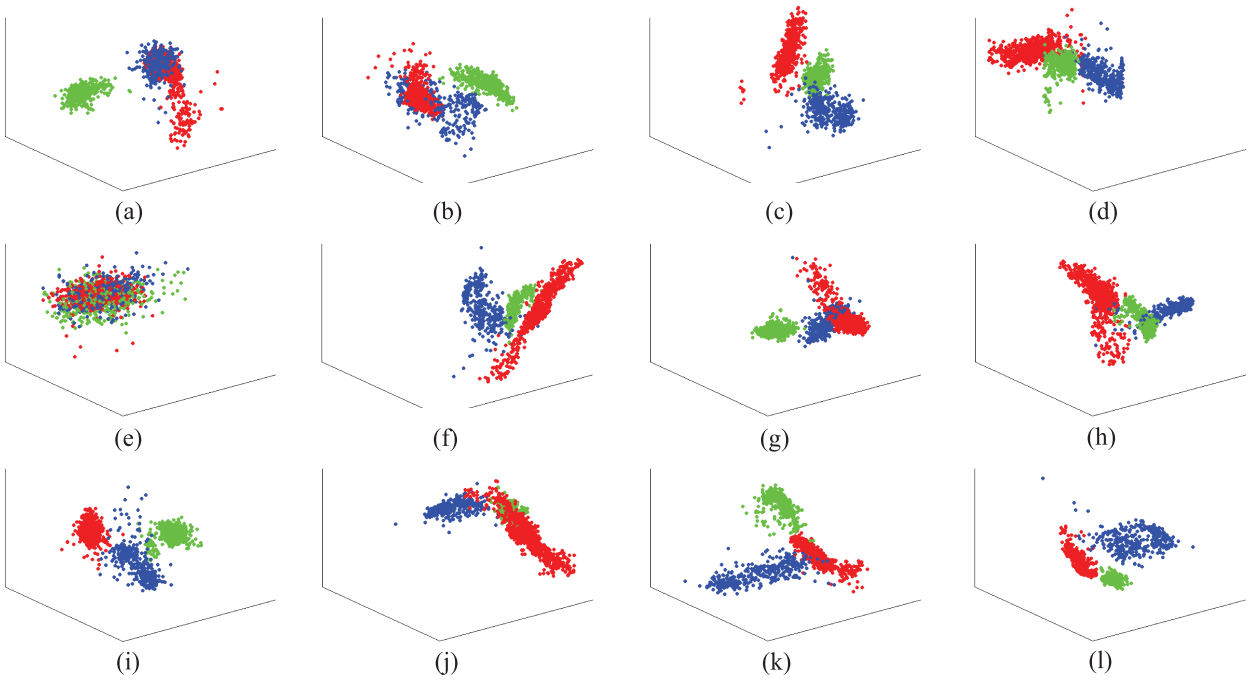


Fig. 8. 3-D embedding of three-class synthetic data set with different DR methods. (a) PCA. (b) NPE. (c) LPP. (d) LGSFA. (e) SPP. (f) DLSP. (g) SLGDA. (h) LGDA. (i) SDME. (j) MSME. (k) DHLP. (l) LC-SMHL.

in data. Manifold-based methods, LPP and LGSFA, and representation-based methods, DLSP, SLGDA, and LGDA, present better than unsupervised methods but still exist some small overlap areas. In comparison with LC-SMHL, the SDME possesses small interclass distance, while DHLP produces lager intraclass distance. Fig. 8(l) demonstrates that the proposed LC-SMHL provides the best intraclass aggregation and interclass separability among all DR methods that bring great help for the subsequent classification task.

In real applications, DR methods normally do not map data into 3-D space for losing too much useful information. To select optimal embedding dimension, we turn the number of dimension with a given set $\{5, 10, 15, 20, 25, 30, 35, 40, 45, 50\}$; 10% samples of the Indian Pines data set, and 2% samples of the Heihe and Salinas data sets were chosen for training, while the remained samples are verified by NN classifier. Fig. 9 exhibits the average OAs of DR methods with different embedding dimensions on three data sets.

It is obvious that the classification performance of each algorithm is unsatisfactory when the embedding dimension is small for losing too much significant discriminant features. Furthermore, we note that the embedding performance of LC-SMHL is obviously superior to other methods even with an extremely low dimension, which further demonstrates that the proposed method can preserve the effective information of original data and enhance the discriminant power significantly. When dealing with unknown samples from HSI data, a small part of samples will be first manually calibrated, and then, the embedding dimension turning experiment will be conducted on the calibration sample set to find the optimal embedding dimension. In order to ensure algorithms achieve satisfactory results in the following experiments, we set embedding dimension $d = 30$ according to the classification curves on three HSI data sets.

TABLE II

CLASSIFICATION RESULTS OF DR METHODS WITH NN ON THE INDIAN PINES DATA SET [OA $\pm$ STD (%)]

Methods	P=2%	P=4%	P=6%	P=8%	P=10%
RSF	57.4±2.1	62.4±0.6	64.7±0.5	66.9±0.4	67.6±0.3
PCA	57.2±2.1	63.3±0.8	65.6±0.5	67.7±0.5	68.2±0.3
NPE	49.0±1.3	64.5±0.6	67.2±0.6	68.4±0.5	69.8±0.1
LPP	47.4±2.0	65.7±1.2	71.2±1.0	75.1±0.9	77.1±0.2
LGSFA	56.4±2.3	66.9±1.4	71.9±0.9	75.4±0.8	77.5±0.6
SPP	52.0±1.3	59.5±1.1	64.5±0.9	67.2±0.7	68.3±0.5
DLSP	54.7±3.9	63.3±1.8	64.7±1.0	67.4±0.9	70.1±0.7
SLGDA	59.0±1.0	64.2±0.7	68.1±0.8	70.8±0.7	72.1±0.7
LGDA	59.8±2.3	65.6±0.7	68.1±0.6	71.1±0.5	72.2±0.5
DBN	61.2±2.9	69.3±2.3	73.3±1.6	74.1±1.7	74.8±1.4
SAE	60.7±2.2	67.1±1.8	71.8±1.5	74.6±1.1	75.1±1.1
SDME	60.0±2.0	66.1±0.9	73.5±0.7	76.5±0.7	77.0±0.6
MSME	56.5±1.2	61.7±1.2	65.7±1.0	68.3±0.9	69.3±0.7
DHLP	60.6±2.6	68.5±1.1	73.0±0.9	74.9±0.8	77.1±0.5
LC-SMHL	**62.3±1.4**	**70.4±0.9**	**75.5±0.9**	**78.2±0.8**	**79.5±0.7**

Note: The best results of a column are marked in bold.

F. Comparison of Classification Results

In general, the classification performance of DR methods is sensitive to the number of training samples generally. To evaluate the performance of LC-SMHL under different training conditions, we randomly selected P% of samples per class for training on the Indian Pines, Heihe, and Salinas data sets. For three data sets that have great difference of sample sizes, the proportion P was turned with sets of $\{2\%, 4\%, 6\%, 8\%, 10\%\}$ on the Indian Pines data set and $\{0.4\%, 0.8\%, 1.2\%, 1.6\%, 2\%\}$ on the Heihe and Salinas data sets, respectively. NN classifier was applied to test samples, and each experiment was repeated ten times to reduce the random error. The average OAs with STDs of algorithms on three HSI data sets are given in Tables II–IV, respectively.

From Tables II–IV, we can find that the OAs of all methods are improved with the increase of training samples since a larger training set generally contains more prior information for training a powerful DR model. The manifold-based

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 08:23:59 UTC from IEEE Xplore. Restrictions apply.

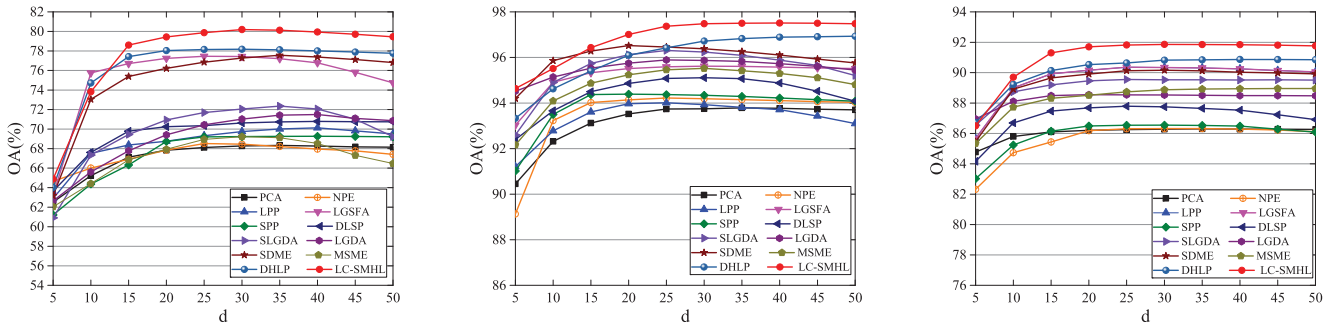


Fig. 9. OAs with different embedding dimensions. (a) Indian Pines. (b) Heihe. (c) Salinas.

TABLE III

CLASSIFICATION RESULTS OF DR METHODS WITH NN ON THE HEIHE DATA SET [OA ± STD (%)]

Methods	P=0.4%	P=0.8%	P=1.2%	P=1.6%	P=2%
RSF	91.3±0.2	92.3±0.1	92.9±0.2	93.4±0.2	93.4±0.2
PCA	91.3±0.2	92.0±0.1	92.9±0.2	93.4±0.2	93.7±0.2
NPE	91.8±0.4	92.3±0.4	93.1±0.2	93.5±0.4	94.1±0.3
LPP	92.2±0.3	92.8±0.1	93.3±0.1	93.7±0.2	93.9±0.3
LGSFA	93.4±0.2	94.6±0.2	95.1±0.1	95.4±0.1	95.6±0.2
SPP	90.8±0.8	92.3±0.5	93.0±0.1	94.0±0.1	94.3±0.2
DLSP	92.2±0.2	93.9±0.1	94.5±0.2	94.9±0.2	95.1±0.2
SLGDA	93.8±0.3	94.7±0.2	95.7±0.2	96.0±0.1	96.2±0.2
LGDA	93.1±0.3	93.7±0.1	94.7±0.1	95.3±0.2	95.8±0.1
DBN	92.2±0.5	93.4±0.2	94.7±0.5	95.3±0.4	95.8±0.3
SAE	92.4±0.6	94.0±0.3	95.1±0.3	95.7±0.2	96.4±0.1
SDME	94.7±0.4	95.7±0.2	96.2±0.2	96.3±0.1	96.3±0.2
MSME	92.8±0.3	93.6±0.2	94.5±0.1	95.2±0.1	95.5±0.2
DHLP	95.2±0.3	95.9±0.3	96.4±0.3	96.6±0.2	96.7±0.1
LC-SMHL	**95.4±0.4**	**96.2±0.3**	**96.8±0.1**	**97.3±0.2**	**97.5±0.2**

TABLE IV

CLASSIFICATION RESULTS OF DR METHODS WITH NN ON THE SALINAS DATA SET [OA ± STD (%)]

Methods	P=0.4%	P=0.8%	P=1.2%	P=1.6%	P=2%
RSF	82.2±0.7	83.8±0.5	84.8±0.4	85.6±0.3	86.0±0.3
PCA	82.6±0.7	83.9±0.4	84.7±0.4	85.9±0.3	86.2±0.3
NPE	79.1±1.1	82.3±0.9	84.4±0.7	85.7±0.6	86.1±0.6
LPP	76.7±0.8	85.5±0.7	88.6±0.6	89.5±0.5	90.2±0.5
LGSFA	82.9±0.6	86.8±0.4	89.5±0.5	90.2±0.3	90.6±0.2
SPP	76.4±0.6	81.8±0.5	84.2±0.5	85.8±0.5	86.4±0.4
DLSP	82.7±1.8	83.8±1.4	84.7±1.1	85.6±0.9	87.3±0.6
SLGDA	83.5±0.6	86.6±0.4	89.3±0.4	89.7±0.3	89.8±0.3
LGDA	82.8±0.6	86.5±0.5	87.4±0.5	88.0±0.4	88.5±0.4
DBN	83.3±2.2	86.1±1.3	87.1±1.7	87.9±0.9	88.3±0.8
SAE	84.4±2.7	86.3±1.2	88.5±1.0	89.2±0.9	89.7±0.6
SDME	83.6±1.2	86.0±0.8	89.5±0.6	89.9±0.5	90.2±0.4
MSME	86.3±1.1	87.5±0.6	88.0±0.4	88.6±0.3	88.8±0.3
DHLP	87.0±0.9	89.9±0.7	89.9±0.4	89.7±0.4	90.2±0.4
LC-SMHL	**87.4±1.2**	**89.1±0.8**	**90.7±0.6**	**91.7±0.5**	**92.2±0.5**

methods, NPE, LPP, and LGSFA, usually produce good classification results with larger training sets but possess unsatisfactory results with less training samples, for the reason that small training set is insufficient to reveal the complex manifold structure embedded in data. With small training sets, DLSP, SLGDA, and SDME generally perform better classification performance than manifold-based methods because they can find the samples from the same subspace and reveal the relationship between samples via sparse coefficients. The deep learning methods DBN and SAE present better classification results than PCA, NPE, DLSP, SLGDA, and LGDA under most conditions because they extract the abstract deep features via the complex networks. However, the data-driven approaches are limited by the finite training samples in HSI.

DHLP and LC-SMHL produce high classification accuracies under the most training conditions due to utilizing hypergraph for revealing the multivariate discriminant relationship between samples. The proposed method LC-SMHL reaches the best classification accuracy in comparison with all DR methods, for the reason that it not only preserves the nonlinear manifold-based structure embedding in the low-dimensional space but also considers the multivariate discriminant sparse relation of HSI data.

For DR methods, they hope to improve the classification performance of most classifiers instead of one certain classifier. Therefore, the universality of DR for different classifiers is a significant evaluating indicator. DBN and SAE are not used in this experiment because they are end-to-end deep learning methods that include a softmax classifier. Besides the above-mentioned experimental results of DR methods combined with the NN classifier, we also combine each algorithm with SVM and SVMCK classifiers for indicating the merits of LC-SMHL. In experiments, 10% samples of the Indian Pines data set and 2% samples of the Heihe and Salinas data sets were randomly chosen as training sets, respectively. The remaining samples as test sets were tested by SVM and SVMCK classifiers. The classification results of DR methods with different classifiers on three HSI data sets are given in Table V.

TABLE V

CLASSIFICATION RESULTS OF DR METHODS COMBINED WITH DIFFERENT CLASSIFIERS ON THREE HSI DATA SETS [OA (%)]

Data sets	Indian Pines		Heihe		Salinas	
Methods	SVM	SVMCK	SVM	SVMCK	SVM	SVMCK
RSF	78.64	92.14	94.08	95.79	88.87	95.09
PCA	78.39	92.43	94.98	96.42	88.42	95.51
NPE	79.34	93.24	95.51	96.61	89.65	95.72
LPP	79.34	95.23	95.42	96.65	91.65	96.47
LGSFA	78.72	96.24	96.47	97.46	90.97	97.09
SPP	76.13	92.39	96.21	96.48	88.79	95.24
DLSP	77.22	92.27	95.85	96.74	91.57	95.91
SLGDA	75.64	96.61	97.28	97.53	91.19	97.37
LGDA	79.44	93.27	96.34	96.94	90.69	95.37
SDME	78.32	96.35	97.11	97.52	90.02	96.26
MSME	77.67	94.04	96.64	96.45	90.36	97.21
DHLP	78.51	95.57	**97.76**	97.27	91.92	97.33
LC-SMHL	**80.41**	**96.82**	97.56	**98.15**	**92.31**	**98.29**

As shown in Table V, DR methods combined SVMCK obtain higher classification accuracies than DR methods

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 03:23:59 UTC from IEEE Xplore. Restrictions apply.

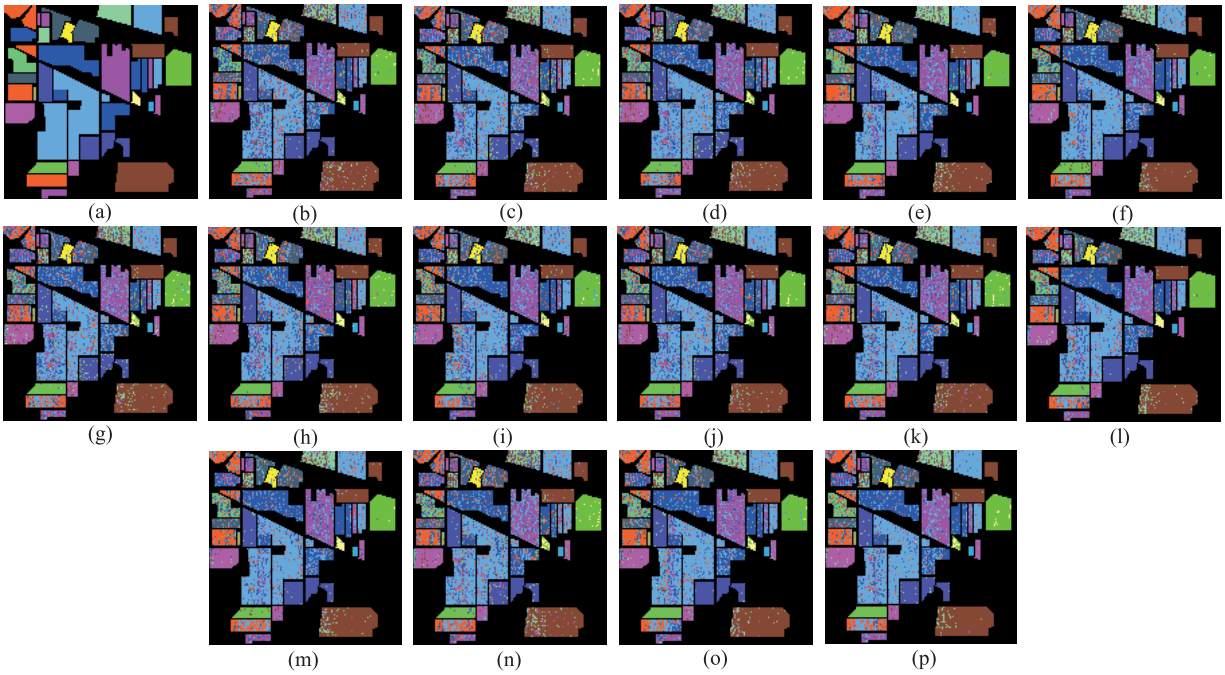


Fig. 10. Classification maps of different DR methods on the Indian Pines data set. (a) GT. (b) RSF (67.7%). (c) PCA (68.2%). (d) NPE (71.4%). (e) LPP (77.2%). (f) LGSFA (77.4%). (g) SPP (68.7%). (h) DLSP (70.6%). (i) SLGDA (72.1%). (j) LGDA (72.5%). (k) DBN (74.5%). (l) SAE (75.3%). (m) SDME (77.8%). (n) MSME (69.2%). (o) DHLP (78.1%). (p) LC-SMHL (79.7%).

TABLE VI
CLASSIFICATION RESULTS OF EACH CLASS IN THE INDIAN PINES DATA SET

Class	RSF	PCA	NPE	LPP	LGSFA	SPP	DLSP	SLGDA	LGDA	DBN	SAE	SDME	MSME	DHLP	LC-SMHL
1	50.3	44.4	44.4	83.3	58.3	41.7	**72.2**	61.1	27.8	52.8	83.3	77.8	38.8	61.1	68.9
2	53.1	53.1	58.4	65.6	69.7	57.5	57.7	69.7	60.5	59.8	68.6	71.1	57.1	70.2	**71.5**
3	49.4	47.8	51.3	**66.7**	61.2	47.7	54.8	51.1	49.3	60.8	55.7	63.6	54.6	62.6	58.6
4	36.2	39.9	45.5	61.5	39.4	33.8	34.3	26.8	44.6	53.6	46.4	47.5	41.7	56.3	**63.9**
5	80.2	81.4	82.1	91.7	88.5	80.2	84.4	81.4	85.1	84.3	85.9	89.7	85.0	91.7	**92.4**
6	95.7	88.4	88.2	96.1	97.4	89.4	93.5	93.6	92.1	94.8	92.6	97.1	91.9	97.2	**97.4**
7	94.4	83.3	83.3	94.4	83.3	88.9	88.9	88.9	88.9	94.4	94.4	83.3	88.8	**94.4**	90.3
8	96.1	96.3	96.7	96.5	**98.8**	93.3	94.2	92.6	96.5	97.7	94.2	99.1	94.8	96.3	98.3
9	70.0	90.0	70.0	90.0	90.0	70.0	70.0	60.0	80.0	70.0	90.0	90.0	50.0	**100**	**100**
10	65.1	65.5	69.3	72.3	68.4	68.3	66.7	67.4	73.3	74.5	58.9	70.9	56.2	66.4	**73.5**
11	67.6	71.3	75.2	76.1	79.9	70.9	68.7	70.9	72.8	79.9	81.5	78.4	71.4	72.4	**82.4**
12	44.8	36.7	47.8	63.1	63.1	39.4	46.1	55.4	47.8	32.3	48.2	70.6	42.3	**64.4**	61.6
13	87.5	91.9	92.4	97.8	90.8	95.1	97.4	88.6	95.7	97.1	94.3	93.5	88.5	97.8	**98.3**
14	90.3	92.2	92.7	91.7	94.5	91.8	90.2	91.4	93.1	95.4	93.1	92.4	88.6	93.6	**95.3**
15	31.1	35.2	37.8	54.8	51.6	34.9	33.4	51.7	44.1	43.6	53.5	55.8	25.9	43.5	**58.5**
16	89.2	89.2	90.4	89.2	88.2	86.8	**91.7**	77.1	85.5	89.2	86.7	89.2	87.9	90.3	89.2
OA	67.7	68.2	71.4	77.2	77.4	68.7	70.6	72.1	72.5	74.6	75.3	77.8	69.2	78.1	**79.7**
AA	68.8	69.2	70.3	78.3	76.4	68.1	71.7	70.4	71.1	75.8	76.7	78.6	67.5	78.9	**80.9**
Kappa	0.63	0.64	0.67	0.74	0.74	0.64	0.65	0.68	0.68	0.71	0.71	0.74	0.65	0.75	**0.77**

Note: The best results of a row are marked in bold.

combined with NN and SVM classifiers; the reason is that NN and SVM are spectral-based classifiers that only consider the spectral features, while SVMCK is spatial–spectral combined classifier that utilizes both spatial and spectral information of HSI. Compared with the RSF of HSI, almost all DR methods present better classification performance for removing the redundant information of original high-dimensional data. Among all DR methods, LC-SMHL achieves the best classification results on almost all data sets no matter combined with any classifiers. Compared with SMDE that also uses manifold-based sparse model, the gains of LC-SMHL over SMDE are 2.09%, 0.45%, and 2.29% on the Indian Pines, Heihe, and Salinas data sets with NN classifier, respectively. The reason for this appearance is that the LCSMC model in LC-SMHL is able to select sparse neighbors from the same manifold and obtain manifold-based sparse coefficients via local spectral constraint and similarity weights matrix. With regard to DHLP that also relies on the hypergraph framework, the proposed LC-SMHL still outperforms DHLP in most experimental conditions, which benefits from the fact that the manifold-based sparse hypergraph of LC-SMHL is able to preserve both local manifold structure and reveal the sparse discriminant relationship between samples.

In order to analyze the classification performance of the LC-SMHL algorithm for different land cover types in HSIs, we randomly choose 10% samples from each class on the Indian

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 03:23:59 UTC from IEEE Xplore. Restrictions apply.



Fig. 11. Classification maps of different DR methods on the Heihe data set. (a) GT. (b) RSF (93.6%). (c) PCA (93.9%). (d) NPE (94.1%). (e) LPP (93.9%). (f) LGSFA (95.6%). (g) SPP (94.4%). (h) DLSP (95.1%). (i) SLGDA (96.2%). (j) LGDA (95.8%). (k) DBN (95.5%). (l) SAE (96.1%). (k) SDME (96.3%). (l) MSME (95.7%). (m) DHLP (96.7%). (n) LC-SMHL (97.5%).

TABLE VII
CLASSIFICATION RESULTS OF EACH CLASS IN THE HEIHE DATA SET

Class	RSF	PCA	NPE	LPP	LGSFA	SPP	DLSP	SLGDA	LGDA	DBN	SAE	SDME	MSME	DHLP	LC-SMHL
1	96.1	95.9	96.3	96.1	97.7	96.9	96.5	97.4	97.1	97.3	97.6	97.7	98.1	98.1	**98.6**
2	95.7	96.4	96.2	96.1	96.6	97.0	96.5	98.3	97.8	98.4	98.2	97.7	98.5	**98.7**	98.5
3	93.1	93.5	93.0	93.4	94.6	94.1	94.1	95.1	95.2	93.9	96.9	96.8	96.7	96.5	**97.6**
4	87.1	87.3	89.7	89.5	90.4	85.7	90.6	93.7	92.3	86.1	88.5	92.2	91.4	91.1	**92.9**
5	90.4	90.9	92.5	90.4	95.5	95.6	95.1	96.3	95.2	93.3	93.9	95.8	96.8	**98.6**	97.5
6	68.3	71.8	65.3	68.1	81.2	55.9	80.2	72.2	81.2	81.2	83.4	70.4	77.3	73.1	**81.4**
7	57.9	53.5	54.6	51.1	66.1	57.1	73.4	72.6	60.6	75.2	70.6	72.7	72.7	68.3	**80.3**
8	80.3	84.4	90.4	86.7	93.9	85.1	96.7	95.9	92.9	90.8	90.1	95.1	96.1	96.5	**97.9**
OA	93.6	93.9	94.1	93.9	95.6	94.4	95.1	96.2	95.8	95.5	96.1	96.3	95.7	96.7	**97.5**
AA	83.6	84.2	84.7	83.9	89.5	83.4	90.4	90.1	89.0	89.5	89.9	89.8	89.1	90.1	**93.1**
Kappa	0.91	0.92	0.92	0.92	0.94	0.92	0.93	0.95	0.94	0.94	0.95	0.95	0.94	0.96	**0.97**

Pine data set and 2% samples from each class on the Heihe and Salinas data sets as training sets. The remaining samples were tested with the NN classifier. Experimental results of different DR methods on the two data sets are given in Tables VI–VIII, while the corresponding classification maps are shown in Figs. 10–12 to visualize the classification results.

According to Tables VI–VIII, the classification results of unsupervised methods, PCA, MSME, and SPP, are inferior to other supervised methods because of ignoring the poor label information. Benefiting from the hypergraph framework, DHLP and LC-SMHL present better classification performance compared with other methods. Among all methods in experiments, the proposed LC-SMHL possesses the best OA, AA, and kappa on the two HSI data sets, and it also achieves the best classification result in ten classes on the Indian Pine data set, six classes on the Heihe data set, and ten classes on the Salinas data set. As shown in Figs. 10–12, the classification maps of LC-SMHL are significantly smoother than other methods. Especially, LC-SMHL produces fewer misclassification pixels and achieve better classification accuracy in

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 03:23:59 UTC from IEEE Xplore. Restrictions apply.

TABLE VIII
CLASSIFICATION RESULTS OF EACH CLASS IN THE SALINAS DATA SET

Class	RSF	PCA	NPE	LPP	LGSFA	SPP	DLSP	SLGDA	LGDA	DBN	SAE	SDME	MSME	DHLP	LC-SMHL
1	97.6	97.5	98.1	99.4	99.5	98.6	98.1	99.6	98.5	99.0	98.9	**99.6**	98.8	99.3	99.5
2	98.9	99.2	99.4	99.9	99.7	98.5	99.4	99.9	98.9	99.5	98.7	99.7	99.2	**99.8**	**99.8**
3	96.1	94.2	98.9	**99.8**	99.1	97.3	96.1	98.4	97.3	78.6	93.8	99.5	99.6	99.7	98.7
4	98.9	99.7	96.5	98.8	99.5	99.6	99.6	98.8	99.6	98.2	99.1	99.1	97.9	99.5	**99.7**
5	96.7	95.5	96.7	99.1	98.4	98.1	97.3	96.6	96.1	96.4	94.6	99.1	99.0	98.9	**99.4**
6	99.3	99.5	99.7	99.8	99.7	99.6	99.6	99.7	99.7	99.5	98.9	99.5	99.6	99.6	**99.8**
7	99.3	99.1	99.6	**100**	99.8	99.3	98.7	**100**	99.4	99.5	99.4	99.9	99.5	99.9	99.9
8	74.9	72.3	70.3	81.2	83.5	71.6	71.2	75.3	73.6	87.4	84.1	**82.4**	80.1	80.3	79.3
9	97.2	97.7	98.4	100	99.9	99.2	98.2	99.8	99.8	98.4	98.7	99.6	99.2	99.6	**99.9**
10	83.5	86.2	84.1	94.2	93.7	88.7	91.4	94.2	92.5	85.6	88.8	93.7	95.5	96.3	**96.5**
11	86.5	89.6	87.8	99.1	**99.7**	93.9	86.6	97.4	93.7	85.4	98.7	99.7	98.8	97.7	98.9
12	99.3	97.5	99.5	99.8	99.1	97.5	98.4	99.8	99.9	98.7	99.8	99.8	99.6	99.8	**100**
13	97.8	96.9	96.2	98.7	99.2	95.8	97.2	99.3	99.6	98.1	98.0	98.6	97.7	**99.1**	98.3
14	89.9	92.4	88.9	92.4	95.3	93.2	92.1	89.5	94.5	88.0	90.1	92.3	96.9	98.3	**99.4**
15	55.4	59.4	61.3	60.6	57.7	50.9	60.4	68.0	64.8	57.0	63.9	61.9	63.1	67.4	**72.5**
16	94.1	92.9	93.2	99.0	98.9	99.1	97.3	97.9	97.9	90.9	95.8	98.9	98.5	99.3	**99.5**
OA	85.9	86.3	86.2	89.8	90.1	85.8	86.7	89.5	88.7	88.5	89.8	90.8	88.3	90.2	**92.4**
AA	91.2	91.8	91.6	94.9	95.1	91.2	92.5	94.6	94.1	91.3	93.8	95.9	93.4	95.1	**97.2**
Kappa	0.84	0.85	0.85	0.89	0.89	0.84	0.88	0.87	0.87	0.87	0.89	0.90	0.87	0.89	**0.91**

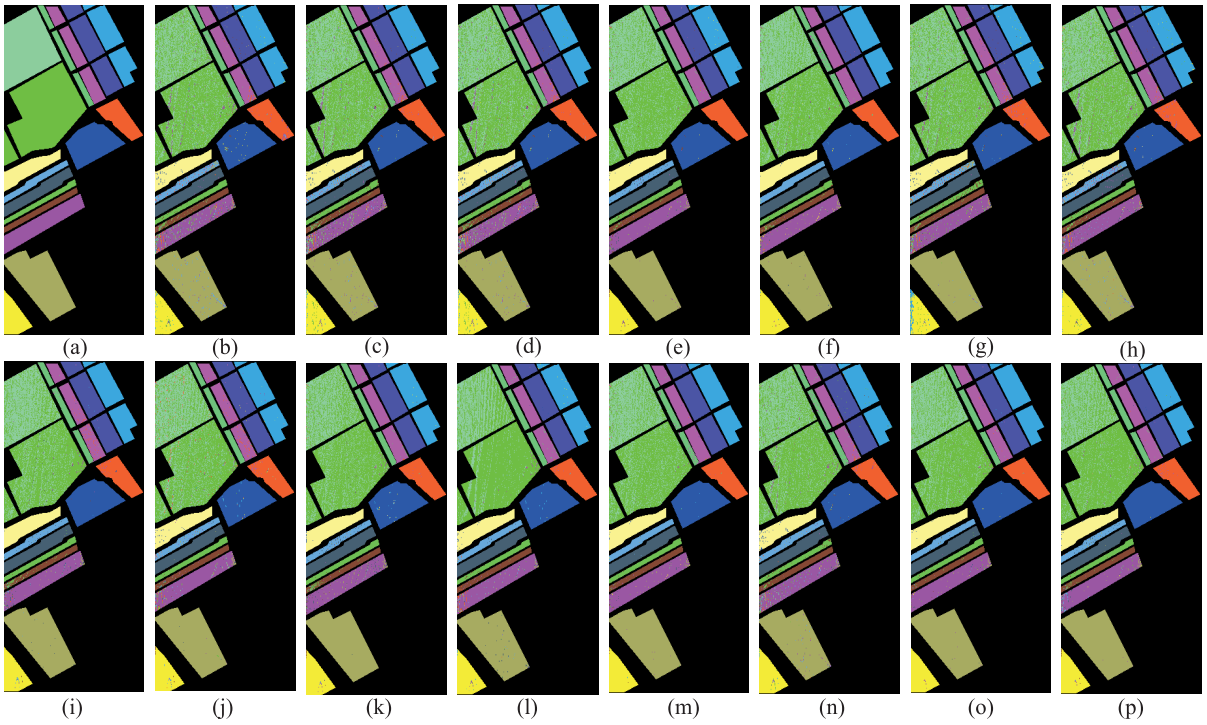


Fig. 12. Classification maps of different DR methods on the Salinas data set. (a) GT. (b) RSF (85.9%). (c) PCA (86.3%). (d) NPE (86.2%). (e) LPP (89.8%). (f) LGSFA (90.1%). (g) SPP (85.8%). (h) DLSP (86.7%). (i) SLGDA (89.5%). (j) LGDA (88.5%). (k) DBN (88.5%). (l) SAE (89.8%). (m) SDME (90.8%). (n) MSME (88.3%). (o) DHLP (90.2%). (p) LC-SMHL (92.4%).

some target areas, such as Soybeans-min, Wheat, and Woods on the Indian Pines data set, Endive Sprout, Watermelon, and Artificial Surface on the Heihe data set, and Lettuce 4wk, Vinyard untrained, and Vinyard trrellis on the Salinas data set. These appearances clearly demonstrate that the proposed method is more effective to enhance the discrimination power of embedding features and weaken the influence of noisy points, which is beneficial for the designed LCSMC SR model and the constructed sparse hypergraphs.

IV. CONCLUSION

In this article, we present a new DR method LC-SMHL based on the manifold-based SR and the hypergraph framework. The proposed method first designs a novel sparse model LCSMC to discover the sparse relation between different local manifold neighborhoods instead of between samples. Meanwhile, LCSMC imposes a local spectral constraint to preserve the local structure of HSI and guarantee that the coefficients are sparse enough. Relying on the LCSMC model and hypergraph framework, LC-SMHL constructs two manifold-based spare hypergraphs to reveal the discriminant multivariate sparse relationship between samples. Then, the objective optimization model of LC-SMHL is built to improve the intraclass aggregation and interclass separability in the embedding process, which enhances the discriminating power of embedding features. As a result, the learned projection

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 03:23:59 UTC from IEEE Xplore. Restrictions apply.

matrix of LC-SMHL is able to extract useful discriminating features for improving HSI classification accuracy significantly.

The proposed approaches have been compared with some state-of-the-art DR methods and SR models on three HSI data sets, and the experimental results have illustrated that the proposed method has excellent DR performance and general applicability for different HSI data sets. Future work will consider introducing spatial information into a manifold-based SR model for extracting the spatial–spectral combined sparse features of HSI.

ACKNOWLEDGMENT

The authors would like to thank Prof. Xue Zhaohui for providing the calibrated Heihe data set and the anonymous reviewers for their comments on this article.

REFERENCES

[1] X. Zheng, Y. Yuan, and X. Lu, "A deep scene representation for aerial scene classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 7, pp. 4799–4809, Jul. 2019.

[2] W. Li, L. Zhang, L. Zhang, and B. Du, "GPU parallel implementation of isometric mapping for hyperspectral classification," *IEEE Geosci. Remote Sens. Lett.*, vol. 14, no. 9, pp. 1532–1536, Sep. 2017.

[3] H. Yang, Q. Du, H. Su, and Y. Sheng, "An efficient method for supervised hyperspectral band selection," *IEEE Geosci. Remote Sens. Lett.*, vol. 8, no. 1, pp. 138–142, Jan. 2011.

[4] X. Zheng, Y. Yuan, and X. Lu, "Hyperspectral image denoising by fusing the selected related bands," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 5, pp. 2596–2609, May 2019.

[5] G. Cavallaro, M. Dalla Mura, J. A. Benediktsson, and L. Bruzzone, "Extended self-dual attribute profiles for the classification of hyperspectral images," *IEEE Geosci. Remote Sens. Lett.*, vol. 12, no. 8, pp. 1690–1694, Aug. 2015.

[6] Q. Shi, L. Zhang, and B. Du, "Semisupervised discriminative locally enhanced alignment for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 51, no. 9, pp. 4800–4815, Sep. 2013.

[7] Y. Dong, B. Du, L. Zhang, and L. Zhang, "Dimensionality reduction and classification of hyperspectral images using ensemble discriminative local metric learning," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 5, pp. 2509–2524, May 2017.

[8] Z. Y. Wang and B. B. He, "Locality perserving projections algorithm for hyperspectral image dimensionality reduction," in *Proc. Int. Conf. Geoinform.*, 2011, pp. 1–4.

[9] X. He, D. Cai, S. Yan, and H.-J. Zhang, "Neighborhood preserving embedding," in *Proc. 10th IEEE Int. Conf. Comput. Vis.*, Beijing, China, Oct. 2005, pp. 1208–1213.

[10] Z. Feng, S. Yang, S. Wang, and L. Jiao, "Discriminative spectral–spatial margin-based semisupervised dimensionality reduction of hyperspectral data," *IEEE Geosci. Remote Sens. Lett.*, vol. 12, no. 2, pp. 224–228, Feb. 2015.

[11] F. Luo, H. Huang, Y. Duan, J. Liu, and Y. Liao, "Local geometric structure feature for dimensionality reduction of hyperspectral imagery," *Remote Sens.*, vol. 9, no. 8, p. 790, Aug. 2017.

[12] X. Zheng, Y. Yuan, and X. Lu, "A target detection method for hyperspectral image based on mixture noise model," *Neurocomputing*, vol. 216, pp. 331–341, Dec. 2016.

[13] X. Zhang, Q. Song, Z. Gao, Y. Zheng, P. Weng, and L. C. Jiao, "Spectral–spatial feature learning using cluster-based group sparse coding for hyperspectral image classification," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 9, no. 9, pp. 4142–4159, Sep. 2016.

[14] D. Liu, S. Li, and L. Fang, "Hyperspectral image classification by exploiting the spectral-spatial correlations in the sparse coefficients," in *Proc. 6th Chin. Conf. Pattern Recognit. (CCPR)*, Nov. 2014, pp. 151–158.

[15] Z. Xue, P. Du, H. Su, and S. Zhou, "Discriminative sparse representation for hyperspectral image classification: A semi-supervised perspective," *Remote Sens.*, vol. 9, no. 4, p. 386, Apr. 2017.

[16] L. Qiao, S. Chen, and X. Tan, "Sparsity preserving projections with applications to face recognition," *Pattern Recognit.*, vol. 43, no. 1, pp. 331–341, Jan. 2010.

[17] F. Zang and J. Zhang, "Discriminative learning by sparse representation for classification," *Neurocomputing*, vol. 74, no. 13, pp. 2176–2183, Jun. 2011.

[18] W. Li, J. Liu, and Q. Du, "Sparse and low-rank graph for discriminant analysis of hyperspectral imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 7, pp. 4094–4105, Jul. 2016.

[19] F. Feng, W. Li, Q. Du, and B. Zhang, "Dimensionality reduction of hyperspectral image with graph-based discriminant analysis considering spectral similarity," *Remote Sens.*, vol. 9, no. 4, p. 323, Mar. 2017.

[20] W. Sun *et al.*, "UL-Isomap based nonlinear dimensionality reduction for hyperspectral imagery classification," *ISPRS J. Photogramm. Remote Sens.*, vol. 89, pp. 25–36, Mar. 2014.

[21] B. Pan, Z. Shi, N. Zhang, and S. Xie, "Hyperspectral image classification based on nonlinear spectral–spatial network," *IEEE Geosci. Remote Sens. Lett.*, vol. 13, no. 12, pp. 1782–1786, Dec. 2016.

[22] Y. Yan Tang, H. Yuan, and L. Li, "Manifold-based sparse representation for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 52, no. 12, pp. 7606–7618, Dec. 2014.

[23] L. Wang, H. Wu, and C. Pan, "Manifold regularized local sparse representation for face recognition," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 25, no. 4, pp. 651–659, Apr. 2015.

[24] E. Elhamifar and R. Vidal, "Sparse manifold clustering and embedding," in *Proc. Adv. Neural Inf. Process. Syst.*, Granada, Spain, Dec. 2011, pp. 55–63.

[25] H. Huang, F. Luo, J. Liu, and Y. Yang, "Dimensionality reduction of hyperspectral images based on sparse discriminant manifold embedding," *ISPRS J. Photogramm. Remote Sens.*, vol. 106, pp. 42–54, Aug. 2015.

[26] W. He, H. Zhang, L. Zhang, W. Philips, and W. Liao, "Weighted sparse graph based dimensionality reduction for hyperspectral images," *IEEE Geosci. Remote Sens. Lett.*, vol. 13, no. 5, pp. 686–690, May 2016.

[27] F. Luo, B. Du, L. Zhang, L. Zhang, and D. Tao, "Feature learning using spatial–spectral hypergraph discriminant analysis for hyperspectral image," *IEEE Trans. Cybern.*, vol. 49, no. 7, pp. 2406–2419, Jul. 2019.

[28] C. Dong, M. Naghedolfeizi, D. Aberra, and X. Zeng, "Spectral–spatial discriminant feature learning for hyperspectral image classification," *Remote Sens.*, vol. 11, no. 13, p. 1552, Jul. 2019.

[29] F. Luo, L. Zhang, B. Du, L. Zhang, and Y. Dong, "Discriminant spatial–spectral hypergraph learning for hyperspectral image classification," in *Proc. IEEE Int. Geosci. Remote Sens. Symp. (IGARSS)*, Jul. 2018, pp. 8480–8483.

[30] S. Huang, D. Yang, Y. Ge, and X. Zhang, "Discriminant hyper-Laplacian projections and its scalable extension for dimensionality reduction," *Neurocomputing*, vol. 173, pp. 145–153, Jan. 2016.

[31] Y. Gan, F. Luo, J. Liu, B. Lei, T. Zhang, and K. Liu, "Feature extraction based multistructure manifold embedding for hyperspectral remote sensing image classification," *IEEE Access*, vol. 5, pp. 25069–25080, 2017.

[32] W. Sun and Q. Du, "Graph-regularized fast and robust principal component analysis for hyperspectral band selection," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 6, pp. 3185–3195, Jun. 2018.

[33] L. Pan, H.-C. Li, Y.-J. Deng, F. Zhang, X.-D. Chen, and Q. Du, "Hyperspectral dimensionality reduction by tensor sparse and low-rank graph-based discriminant analysis," *Remote Sens.*, vol. 9, no. 5, p. 452, May 2017.

[34] G. Camps-Valls, L. Gomez-Chova, J. Munoz-Mari, J. Vila-Frances, and J. Calpe-Maravilla, "Composite kernels for hyperspectral image classification," *IEEE Geosci. Remote Sens. Lett.*, vol. 3, no. 1, pp. 93–97, Jan. 2006.

[35] Y. Chen, N. M. Nasrabadi, and T. D. Tran, "Hyperspectral image classification using dictionary-based sparse representation," *IEEE Trans. Geosci. Remote Sens.*, vol. 49, no. 10, pp. 3973–3985, Oct. 2011.

[36] T. Saha, S. Srivastava, S. Khare, P. S. Stanimirović, and M. D. Petković, "An improved algorithm for basis pursuit problem and its applications," *Appl. Math. Comput.*, vol. 355, pp. 385–398, Aug. 2019.

[37] Z. Xue, P. Du, J. Li, and H. Su, "Sparse graph regularization for robust crop mapping using hyperspectral remotely sensed imagery with very few *in situ* data," *ISPRS J. Photogramm. Remote Sens.*, vol. 124, pp. 1–15, Feb. 2017.

[38] Y. Chen, Z. Lin, X. Zhao, G. Wang, and Y. Gu, "Deep learning-based classification of hyperspectral data," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 7, no. 6, pp. 2094–2107, Jun. 2014.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 03:23:59 UTC from IEEE Xplore. Restrictions apply.



Yule Duan received the B.S. degree in measuring and testing technology and instruments from Tianjin Polytechnic University, Tianjin, China, in 2016. He is pursuing the Ph.D. degree in instrument science and technology with Chongqing University, Chongqing, China.

His research interests are pattern recognition, machine learning, and image processing in general.



Hong Huang (Member, IEEE) received the M.S. degree in pattern recognition and the Ph.D. degree in instrument science and technology from Chongqing University, Chongqing, China, in 2005 and 2008, respectively.

He is a Professor with the Image Information Processing Laboratory, Chongqing University, and a Visiting Scholar with the Department of Electrical, Computer, and Biomedical Engineering, University of Rhode Island, Kingston, RI, USA. His main research activities are in the fields of image processing, pattern recognition, and remote sensing. In particular, he pays more attention to sparse representation, manifold learning, and hyperspectral remote sensing.

Dr. Huang is a Reviewer of over 20 international journals, including the IEEE TRANSACTIONS ON GEOSCIENCE AND REMOTE SENSING, the *ISPRS Journal of Photogrammetry and Remote Sensing*, and the IEEE TRANSACTIONS ON MEDICAL IMAGING, *Neurocomputing*, and *Information sciences*.



Yuxiao Tang received the B.S. degree in instrument science and technology from the Hefei University of Technology, Anhui, China, in 2017. He is pursuing the master's degree in instrument science and technology with Chongqing University, Chongqing, China.

His research interests are hyperspectral image feature extraction, image processing, and face recognition.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 03:23:59 UTC from IEEE Xplore. Restrictions apply.